Concentration Spectrum of the Half-Filled Sine Kernel: Periods, Tails and Plunge Windows

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Scope

Five articles on the discrete concentration problem for the half-filled sine kernel: higher-order modular periods, concentration eigenvalues from logarithmic connection coefficients, spectral tails and exact concentration logits, reciprocal traces and the plunge law, and positive moment bounds on growing plunge windows. They supply reference-kernel inputs used in the Wilson–Dirac relative-entropy papers.

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Higher-Order Modular Periods for the Half-Filled Sine Kernel

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Abstract

The half-filled finite sine kernel commutes with a Jacobi matrix whose eigenvector generating function satisfies an exact differential equation. We derive its formal even WKB periods through order N^{-5} and reduce each correction to two derivatives of a hypergeometric period. Finite polynomial identities certify the reductions. Their edge poles reproduce three shifted-Stirling coefficients exactly at the level of formal period functions. Separately, a positive Fourier-integral formula for the concentration eigenvalues gives an explicit Gauss–Legendre truncation bound and a useful additional numerical check. Multiprecision comparisons at thirteen bulk modes support the proposed spectral expansion. The separate companion *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* identifies these formal periods with the spectral coefficients and proves a uniform remainder on each fixed interior interval. The present paper supplies the coefficient reductions. These results concern one specified reference model; they do not establish the completeness of a physical model or a quantum-field-theoretic mass gap.

1 Model, conventions and scope

This study develops the reference-modular discrepancy used in *From Prephysical Hypotheses to Physical Models*. The mathematical objects and evidence levels are explicit: finite identities and formal coefficient reductions are proved below; comparisons with the concentration spectrum are numerical evidence. The established commuting-operator approach to the free-fermion entanglement Hamiltonian provides the setting [1]. No priority claim for the classical ingredients or the displayed higher-order formulae is made.

For even N, indices $0 \le j, k < N$, define

$$B_{jk} = K(j-k), \qquad K(0) = \tfrac{1}{2}, \qquad K(d) = \frac{\sin(\pi d/2)}{\pi d} \quad (d \neq 0). \tag{1}$$

Let J have zero diagonal and positive bonds

$$J_{j-1,j} = J_{j,j-1} = a_j = \frac{j(N-j)}{N}, \qquad 1 \le j < N. \tag{2}$$

The Fourier representation

$$v^* B v = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} \left| \sum_{j=0}^{N-1} v_j e^{ij\theta} \right|^2 \mathrm{d}\theta \tag{3}$$

and its complementary-band counterpart give $0 < B < I$: a nonzero finite Fourier polynomial cannot vanish on an interval. Direct computation, with $d = j - k$, gives

$$[J, B]_{jk} = \frac{N - j - k - 1}{N} \big[(d-1)K(d-1) + (d+1)K(d+1) \big] = 0.$$

Here $0K(0) = 0$ makes the same identity valid when $d = \pm 1$. Because J is an irreducible real Jacobi matrix, its spectrum is simple; its real normalized eigenvectors are also eigenvectors of B. For even N, the maximum absolute row sum is $N/2 - 1/N$, so every eigenvalue satisfies $|\lambda/N| < 1/2$.

Write $Jv=\lambda v$, $Bv=\zeta v$ and

$$t=\lambda/N, \qquad \varepsilon=\log\frac{1-\zeta}{\zeta}, \qquad e=\varepsilon+N\eta(t), \tag{4}$$

$$\eta(t)=\pi t\,{}_3F_2\!\left(\tfrac14,\tfrac12,\tfrac34;1,\tfrac32;4t^2\right), \qquad |t|<\tfrac12. \tag{5}$$

The alternating-sign diagonal matrix $D_{jj}=(-1)^j$ obeys $DJD=-J$ and $DBD=I-B$. Thus the paired mode has $(-t,1-\zeta,-\varepsilon,-e)$. All signs below use (4). In particular, signs of individual corrections cannot be compared without specifying the branch.

2 The exact differential equation and the formal period

Proposition 1 (Finite Fourier equation)**.** *For an eigenvector of* (2)*, put*

$$U(\xi)=\sum_{j=0}^{N-1} v_j e^{i[j-(N-1)/2]\xi}, \qquad p=\cos\xi.$$

Then

$$(pU')'+\left[\frac{N^2-1}{4}p-\frac{N\lambda}{2}\right]U=0. \tag{6}$$

On a region with a chosen square-root branch, $V=\sqrt{p}\,U$ *satisfies*

$$V''+[N^2q(p,t)+r(p)]V=0, \qquad q=\frac{p-2t}{4p}, \qquad r=\frac{1}{4p^2}. \tag{7}$$

Primes in this section denote ξ *derivatives.*

Proof. For $m=j-(N-1)/2$, $Na_{j+1}=(N^2-1)/4-m^2-m$ and $Na_j=(N^2-1)/4-m^2+m$. Multiplying the finite recurrence by $e^{im\xi}$ and summing gives (6); the endpoint bonds vanish. Substitution of $U=p^{-1/2}V$, using $p''=-p$ and $(p')^2=1-p^2$, gives (7). In the alternative notation $N^2/4-\mu/p+1/(4p^2)$ this fixes $\mu=N\lambda/2$. □

For a formal ansatz $V=\exp(i\int Y\,\mathrm{d}\xi)$, the Riccati equation is $Y^2-iY'=N^2q+r$. Its even momentum W, defined by $Y=W+iW'/(2W)$, obeys

$$W^2+\tfrac12(\log W)''-\tfrac14[(\log W)']^2=N^2q+r. \tag{8}$$

The word *formal* means equality of coefficients in an inverse-power series; convergence or a spectral error bound is not assumed. Write

$$W=N\sqrt{q}\left(1+\frac{a}{N^2}+\frac{b}{N^4}+\frac{c}{N^6}+\cdots\right). \tag{9}$$

Expansion of (8) determines the rational functions

$$R_0=\tfrac14(\log q)''-\tfrac{1}{16}[(\log q)']^2, \qquad a=\frac{r-R_0}{2q},$$

$$R_1=\tfrac12 a''-\tfrac14(\log q)'a', \qquad b=-\frac{qa^2+R_1}{2q},$$

$$L_2=b-\tfrac12 a^2, \qquad R_2=\tfrac12 L_2''-\tfrac14(\log q)'L_2'-\tfrac14(a')^2, \qquad c=-\frac{2qab+R_2}{2q}. \tag{10}$$

All derivatives can be evaluated rationally using $f''=(1-p^2)f_{pp}-pf_p$ and $f'g'=(1-p^2)f_pg_p$. For example,

$$a=-\frac{2p^3t+p^2t^2-p^2-t^2}{2p(p-2t)^3}. \tag{11}$$

For $0 < |t| < 1/2$, choose a counterclockwise cycle Γ enclosing the whole cut joining 0 and $2t$, excluding $-1, 1$. Set $\mathrm{d}\xi = -\mathrm{d}p/\sqrt{1-p^2}$, with $\sqrt{1-p^2}$ positive on $(-1,1)$ and $\sqrt{q} \sim 1/2$ outside its two branch points. Both square roots return to their initial values along the cycle. A circle centered at zero with $2|t| < R < 1$ makes the Laurent expansion immediate and gives

$$-i \oint_\Gamma \sqrt{q}\, \mathrm{d}\xi = \eta(t). \tag{12}$$

Indeed, expanding $(1-2t/p)^{1/2}(1-p^2)^{-1/2}$ and taking the coefficient of p^{-1} yields (5), with its leading term πt. This fixes the normalization without any concentration data. Define

$$P_{\text{form}} = -i \oint_\Gamma W\, \mathrm{d}\xi = N\eta - \frac{\varphi_1}{N} - \frac{\varphi_3}{N^3} - \frac{\varphi_5}{N^5} - \cdots, \tag{13}$$

$$\varphi_1 = i \oint_\Gamma \sqrt{q}\, a\, \mathrm{d}\xi, \qquad \varphi_3 = i \oint_\Gamma \sqrt{q}\, b\, \mathrm{d}\xi, \qquad \varphi_5 = i \oint_\Gamma \sqrt{q}\, c\, \mathrm{d}\xi. \tag{14}$$

No equality between P_{form} and $\log[\zeta/(1-\zeta)]$ has yet been established by this construction.

3 Exact reduction of the three correction periods

Theorem 2 (Formal coefficient identities)**.** *For $0 < |t| < 1/2$, the period functions in (14) satisfy*

$$\varphi_1 = \frac{t}{2}\eta' + (t^2 - \tfrac{1}{6})\eta'', \qquad \varphi_3 = A_3\eta' + B_3\eta'', \qquad \varphi_5 = A_5\eta' + B_5\eta'', \tag{15}$$

where primes now denote t derivatives and

$$A_3 = \frac{108t^4 - 83t^2 + 7}{120t(4t^2-1)^2}, \qquad B_3 = \frac{708t^6 - 789t^4 + 153t^2 - 14}{360t^2(4t^2-1)^2}, \tag{16}$$

$$A_5 = -\frac{6384t^{10} - 51320t^8 + 26567t^6 - 13594t^4 + 2019t^2 - 124}{1680t^3(4t^2-1)^4}, \tag{17}$$

$$B_5 = -\frac{38304t^{12} - 374688t^{10} + 267522t^8 - 170573t^6 + 39472t^4 - 4937t^2 + 248}{5040t^4(4t^2-1)^4}. \tag{18}$$

Each φ_k is odd in t. These are exact identities between the defined formal period coefficients, independent of a spectral asymptotic assertion.

Proof. Define

$$\mathcal{L}(R) = (1-p^2)R_p - pR + \frac{(1-p^2)t}{p(p-2t)}R.$$

For $f = a, b, c$, the rational identity

$$f = \frac{A}{p-2t} + \frac{B}{(p-2t)^2} + \mathcal{L}(R_f) \tag{19}$$

reduces the integral, because $\sqrt{q}\,\mathcal{L}(R_f) = \mathrm{d}[p'\sqrt{q}\,R_f]/\mathrm{d}\xi$. The primitive is single-valued along Γ, so its closed integral vanishes. Differentiating (12) on a fixed admissible contour gives

$$\eta' = i \oint_\Gamma \frac{\sqrt{q}}{p-2t}\, \mathrm{d}\xi, \qquad \eta'' = i \oint_\Gamma \frac{\sqrt{q}}{(p-2t)^2}\, \mathrm{d}\xi.$$

For completeness the following finite coefficient system certifies (19), including its longer primitives. Set $(f_1, f_2, f_3) = (a, b, c)$, $D_r = p^r(p-2t)^{3r}$ and

$$R_{f_r} = \frac{P_r(p)}{p^{r-1}(p-2t)^{3r-1}}, \qquad \deg P_r \le 4r - 3.$$

Equation (19) is equivalent to the polynomial identity

$$(1-p^2)p(p-2t)P_r' + [-p^2(p-2t) + (2r-1)(1-p^2)(t-2p)]P_r \\ + Ap^r(p-2t)^{3r-1} + Bp^r(p-2t)^{3r-2} = D_r f_r. \tag{20}$$

The degree-$4r$ coefficient cancels. In ascending powers $p^0, \dots, p^{4r-1}$, with unknowns the ascending coefficients of P_r followed by A, B, the coefficient matrices have determinants

$$\det M_1 = -3t^2(4t^2-1), \quad \det M_2 = -2835t^6(4t^2-1)^4, \quad \det M_3 = -30405375t^{10}(4t^2-1)^7. \tag{21}$$

They are nonzero in the stated domain. Exact elimination in these 4, 8 and 12 dimensional systems gives $A = t/2$, $B = t^2 - 1/6$ for $r = 1$, and precisely (16)–(18) for $r = 2, 3$. Substitution makes all 4, 8 and 12 coefficient residuals zero. Thus (10) and (20) are a finite, reproducible rational certificate, requiring no numerical fitting. At first order, $R_a = (-4p+3t)/[6(p-2t)^2]$. Finally, η' is even, η'' odd, all A are odd and all B even, proving oddness. □

The combinations extend through $t = 0$ by cancellation; the displayed rational formulae are applied here only for $0 < |t| < 1/2$. A fixed circle $|p| = R < 1$ and the coefficient recursion give their analytic continuation for $|2t| < R$, which also explains the removability without division by small numerical quantities.

4 Edge poles and coefficient-level Stirling matching

Theorem 3 (Edge limits of the formal functions)**.** *Let $u = 1/2 - t \downarrow 0$ on the positive branch. Then*

$$\lim u\varphi_1 = \frac{\sqrt{2}}{24}, \qquad \lim u^3\varphi_3 = -\frac{14\sqrt{2}}{2880}, \qquad \lim u^5\varphi_5 = \frac{124\sqrt{2}}{40320}. \tag{22}$$

On the negative branch the signs reverse when $u = 1/2 - |t|$.

Proof. Expansion and beta integration give

$$\eta(t) = 4t\int_0^1 \frac{\sqrt{1-s^2}}{\sqrt{1-4t^2s^4}}\,\mathrm{d}s, \\ \eta'(t) = 4\int_0^1 \frac{\sqrt{1-s^2}}{(1-4t^2s^4)^{3/2}}\,\mathrm{d}s, \qquad \eta''(t) = 48t\int_0^1 \frac{s^4\sqrt{1-s^2}}{(1-4t^2s^4)^{5/2}}\,\mathrm{d}s. \tag{23}$$

Differentiation is justified on compact subintervals of $|2t| < 1$. In the last integral put $t = 1/2 - u$ and $s = 1 - uz$. The integrand of $u\eta''$, extended by zero for $z > 1/u$, tends to

$$\frac{3\sqrt{2}}{4}\frac{z^{1/2}}{(1+z)^{5/2}}.$$

For $0 < u < 1/4$, $\sqrt{1-s^2} \le \sqrt{2uz}$ and $1 - 4t^2s^4 \ge c_0 u(1+z)$ with an absolute $c_0 > 0$. The rescaled integrand is therefore bounded by a fixed multiple of the displayed integrable function. Dominated convergence yields

$$u\eta'' \longrightarrow \frac{3\sqrt{2}}{4}\mathrm{B}(3/2, 1) = \frac{1}{\sqrt{2}}.$$

Splitting the integral for η' at $1 - s = u$ gives $\eta' = O(\log(1/u))$. The exact rational coefficients give

$$B_1 \to \tfrac{1}{12}, \qquad u^2B_3 \to -\tfrac{7}{720}, \qquad u^4B_5 \to \tfrac{31}{5040}.$$

The $A\eta'$ terms vanish after multiplication by u, u^3, u^5, respectively. This proves (22); no asymptotic expansion was differentiated without control. □

For the formal comparison variable $h = Nu/\sqrt{2}$, these leading poles have the coefficients $1/(24h)$, $-7/(2880h^3)$ and $31/(40320h^5)$. They coincide with the shifted-Stirling expansion

$$C(h) := h\log h - h + \tfrac{1}{2}\log(2\pi) - \log\Gamma(h+\tfrac{1}{2}) \sim \frac{1}{24h} - \frac{7}{2880h^3} + \frac{31}{40320h^5} - \cdots. \tag{24}$$

This classical Gamma statement follows from duplication and Stirling's formula [3, 2]. For example, if $S(h) = \log\Gamma(h) - (h-1/2)\log h + h - \frac{1}{2}\log(2\pi)$, duplication gives $C(h) = S(h) - S(2h)$. The ordinary first-omitted-term remainder bound for positive h gives the conservative inequality

$$\left|C(h) - \frac{1}{24h} + \frac{7}{2880h^3} - \frac{31}{40320h^5}\right| \le \frac{129}{215040h^7}, \qquad h > 0. \tag{25}$$

This is a Gamma-function bound, not a bound for e. In particular, substituting fixed h into an unproved or nonuniform bulk expansion cannot establish a spectral edge theorem. The exact fifth limit is $124\sqrt{2}/40320 = 0.00434926779001553518366\ldots$ in the declared u convention.

5 A positive computation of tiny concentration eigenvalues

Let $N = 2m$. Reversal symmetry splits J into two m dimensional Jacobi matrices. For parity $\sigma = \pm 1$, the isometry

$$v = \frac{1}{\sqrt{2}}(w, \sigma\ \mathrm{reverse}(w)) \tag{26}$$

gives the same first $m-1$ bonds and final diagonal entry $\sigma a_m = \sigma N/4$. The two matrices differ by a positive rank-one perturbation and strictly interlace, since the final coordinate is cyclic. For the full eigenvalues ordered increasingly from index zero, $\sigma = (-1)^{\mathrm{index}+1}$.

For real normalized w, define

$$f_+(\theta) = \sum_{j=0}^{m-1} w_j \cos((m-\tfrac{1}{2}-j)\theta), \qquad f_-(\theta) = \sum_{j=0}^{m-1} w_j \sin((m-\tfrac{1}{2}-j)\theta).$$

Equation (3) becomes the nonnegative integral

$$\zeta = \frac{2}{\pi}\int_0^{\pi/2} f_\sigma(\theta)^2\,\mathrm{d}\theta. \tag{27}$$

For Q point Gauss–Legendre nodes x_ℓ and weights ω_ℓ on $[-1,1]$, set

$$\zeta_Q = \tfrac{1}{2}\sum_{\ell=1}^{Q} \omega_\ell f_\sigma\left(\frac{\pi(x_\ell+1)}{4}\right)^2. \tag{28}$$

The amplitude can be evaluated in $O(N)$ operations per node by a trigonometric Clenshaw recurrence. With $c_r = w_{m-1-r}$, $x = \cos\theta$, $d_m = d_{m+1} = 0$ and $d_r = c_r + 2xd_{r+1} - d_{r+2}$, $f_+ = \cos(\theta/2)(d_0 - d_1)$ and $f_- = \sin(\theta/2)(d_0 + d_1)$.

Proposition 4 (Exact-arithmetic quadrature bound)**.** *For the exact normalized real vector in* (26)*,*

$$|\zeta - \zeta_Q| \le \frac{N}{2}\left[\frac{\pi(N-1)}{2}\right]^{2Q} \frac{(Q!)^4}{(2Q+1)((2Q)!)^3}. \tag{29}$$

Proof. The Fourier identity gives $f_\sigma^2 = \frac{1}{2}\sum_{j,k} v_j v_k e^{i(j-k)\theta}$. Consequently

$$\sup_\theta |(f_\sigma^2)^{(2Q)}(\theta)| \le \tfrac{1}{2}(\textstyle\sum_j |v_j|)^2(N-1)^{2Q} \le \frac{N}{2}(N-1)^{2Q}.$$

The classical Gauss–Legendre error formula [4], mapped to $[0,\pi/2]$ and multiplied by $2/\pi$, yields (29). □

This bound controls truncation for exact nodes, weights and vector. It does not enclose floating-point or eigenvector errors. Positivity removes the cancellation between order-one terms of the Toeplitz quadratic form, but the trigonometric amplitude itself can still cancel and requires adequate precision.

For $N = 1280$, mode index 412, a separate parity-reduced calculation at 450 decimal digits with $Q = 1280$ gives

$$\zeta = 1.943646181331436383656529372586943984855983487605 9\ldots \times 10^{-219}.$$

Its relative difference from the multiprecision Toeplitz quadratic form is approximately 2.25×10^{-233}. The logarithm to base ten of the right side of (29) is approximately -698.02; the full eigenvector residual norm is approximately 3.92×10^{-449}. These decimal results are numerical checks, not outward-rounded enclosures. A second check at $N = 320$, target $t = -0.35$, finds an occupation of order 10^{-158}; positive quadrature at 200 and 300 digits agrees in all 100 stored significant digits, while the 200 digit Toeplitz calculation loses much more precision through cancellation.

6 Bulk comparisons and the remaining spectral question

To test the formal coefficients independently of their derivation, eigenpairs were selected nearest to targets $t = -0.05, -0.125, -0.25, -0.4$ for $N = 160, 320, 640$, with one further $N = 1280$ mode. Each eigenvalue and eigenvector was recomputed from the finite Jacobi recurrence in multiprecision. Double precision was used only for starting guesses and mode selection. Residual norms and multiprecision Sturm counts on either side of each root were checked. Every concentration calculation was repeated with 70 additional working digits; all 90 stored significant digits of e agreed.

The formal coefficients were evaluated by contour quadrature at two radii, with 1024 and 2048 nodes and 85 and 105 working digits, respectively. Comparisons were made on the 60 significant digits retained by that calculation; agreement beyond the stored resolution is not an exact-zero discrepancy. The first coefficient also agreed with its derivative formula. All corrections in Table 1 use the actual $t = \lambda/N$, not the target.

Define the diagnostics

$$r_5 = e - \frac{\varphi_1(t)}{N} - \frac{\varphi_3(t)}{N^3} - \frac{\varphi_5(t)}{N^5}, \qquad R_7 = N^7 r_5. \tag{30}$$

N	Target t	Actual t	R_7	$\lvert r_5/e\rvert$
160	-0.050	-0.04909225	0.5489811	8.423×10^{-13}
160	-0.125	-0.12510710	2.895189	1.670×10^{-12}
160	-0.250	-0.25111340	67.30762	1.618×10^{-11}
160	-0.400	-0.40225408	63500.15	5.170×10^{-9}
320	-0.050	-0.05018619	0.5648435	1.324×10^{-14}
320	-0.125	-0.12368986	2.810152	2.564×10^{-14}
320	-0.250	-0.24937164	63.9315	2.427×10^{-13}
320	-0.400	-0.40020989	55138.37	7.157×10^{-11}
640	-0.050	-0.04962141	0.5567286	2.063×10^{-16}
640	-0.125	-0.12438918	2.852566	4.042×10^{-16}
640	-0.250	-0.25023632	65.63259	3.872×10^{-15}
640	-0.400	-0.39919070	51341.78	1.052×10^{-12}
1280	-0.125	-0.12473999	2.874033	6.343×10^{-18}

Table 1: Bulk diagnostics evaluated at the actual Jacobi mode coordinates. These finite samples support the formal expansion but do not establish a uniform remainder.

At $N = 1280$ and increasing mode index 412,

$$\begin{aligned} t &= -0.12473999119693865638438797411190629 3\ldots, \\ e &= -0.00008048751355147794775611390542947613274\ldots, \\ \varphi_1 &= -0.10302409444857268246635282815793573 0\ldots, \\ \varphi_3 &= +0.12632528385192875717492837499131009 8\ldots, \\ \varphi_5 &= -0.41197754642447173452223841796269928 7\ldots. \end{aligned}$$

The difference $N^3(e - \varphi_1/N) - \varphi_3$ is approximately -2.51450×10^{-7}. After subtracting φ_3/N^3, the unscaled error is approximately -1.19901×10^{-16}; subtracting φ_5/N^5 leaves $+5.10531 \times 10^{-22}$, or 6.34298×10^{-18} relative to e. The fifth coefficient is negative at this mode; its positive symmetry partner has the opposite sign.

The diagnostics are consistent with a next inverse power N^{-7} for the sampled interior modes. The near-edge target has much larger R_7 than the central targets, underscoring the need for a stated domain and uniform constants. Since the actual coordinates drift with N, this table is not a fixed-t convergence-rate proof.

Precision and structural checks. At fixed actual t, error propagation in the fifth-order diagnostic is

$$\delta[N^5(e - \varphi_1/N - \varphi_3/N^3)] = N^5\delta e - N^4\delta\varphi_1 - N^2\delta\varphi_3.$$

At $N = 1280$, an absolute error 10^{-13} in e alone becomes approximately 343.6 in this diagnostic. Thus binary64 residual data cannot resolve this correction. The correct signed kernel (1) is even in d: either both numerator and denominator use d, or both use $|d|$. Mixing these conventions produces an odd off-diagonal kernel. For a real-vector quadratic form, the resulting matrix $B_{\text{bad}} = I/2 + A$, $A^T = -A$, returns $1/2$ exactly, hence $\varepsilon = 0$ and $e = N\eta$. Hermiticity and explicit signed-offset tests detect this failure.

Spectral connection and remaining edge question. The separate whole-line connection theorem establishes, for every compact $K \subset (-1/2, 0)$ or $K \subset (0, 1/2)$, constants C_K, N_K such that every paired mode with $t \in K$ and even $N \geq N_K$ obeys

$$\left| e - \frac{\varphi_1(t)}{N} - \frac{\varphi_3(t)}{N^3} - \frac{\varphi_5(t)}{N^5} \right| \leq C_K N^{-7}. \tag{31}$$

This inequality is proved in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*, including continuation across zero, through exact boundary normalization and an integrable Riccati residual. It does not follow from the coefficient identities alone. An edge-overlap theorem would require additional uniform estimates as K approaches $\pm 1/2$. Even (31) alone would not control the full modular-discrepancy operator norm or complete the associated Wilson relative-entropy asymptotic, which also involves edge modes and weighted traces. The present model-level results therefore refine a specific mathematical component without establishing a universal physical theory.

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Concentration Eigenvalues from Logarithmic Connection Coefficients

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Abstract

For a finite sine kernel at arbitrary filling, we express each concentration odds ratio exactly as a ratio of derivatives of logarithmic connection coefficients for complementary Sturm–Liouville equations. A Lagrange identity identifies those derivatives with positive norming integrals. The formula includes the parity factor $(-1)^{N-1}$; the familiar minus sign at half filling therefore requires even matrix size. Dividing out the finite Jacobi characteristic polynomial gives an equivalent formula without derivatives. We also give quantitative criteria converting relative connection or solution-norm errors into concentration-logit errors. These are finite identities and sufficient error criteria. Identification with a formal quantum period, including a uniform spectral remainder, is supplied at half filling on fixed interior intervals by the separate companion *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*. A general-filling uniform remainder remains a separate problem.

1 Finite model and conventions

Let $N \geq 1$, $0 < \theta < \pi$, $m_j = j - (N-1)/2$ for $0 \leq j < N$, and $\alpha = (N^2-1)/4$. Define the real symmetric matrices

$$(B_\theta)_{jk} = \begin{cases} \theta/\pi, & j = k, \\ \sin[\theta(j-k)]/[\pi(j-k)], & j \neq k, \end{cases} \tag{1}$$

$$(H_\theta)_{jj} = \cos\theta\, m_j^2, \qquad (H_\theta)_{j-1,j} = (H_\theta)_{j,j-1} = \frac{j(N-j)}{2}. \tag{2}$$

Here $\theta = 2\pi W$ in the bandwidth convention $0 < W < 1/2$. The concentration formulation and its commuting tridiagonal operator are classical; see Slepian [1] and the explicit free-fermion formulation in Eisler–Peschel [2]. Our diagonal convention differs from a frequently used one by a scalar shift. All normalizations needed below are proved directly.

For $U_v(x) = \sum_j v_j e^{i m_j x}$, Parseval's identity gives

$$v^* B_\theta v = \frac{1}{2\pi}\int_{-\theta}^{\theta} |U_v(x)|^2\, \mathrm{d}x, \qquad \|v\|^2 = \frac{1}{2\pi}\int_{-\theta}^{2\pi-\theta} |U_v(x)|^2\, \mathrm{d}x. \tag{3}$$

Consequently $0 < B_\theta < I$: a nonzero finite Fourier polynomial cannot vanish on an open interval. A concentration eigenvalue is denoted ζ, and its logit is

$$\mathcal{P} = \log\frac{\zeta}{1-\zeta} = -\varepsilon. \tag{4}$$

All logarithms are natural. No physical time or energy scale is assigned to the differential-equation parameters by these definitions.

Lemma 1 (Exact Fourier equation and commutation)**.** *Put* $p_\theta(x) = \cos x - \cos\theta$ *and* $\mathcal{L}_\theta U = (p_\theta U')' + \alpha \cos x\, U$*. Then* $\mathcal{L}_\theta U_v = U_{H_\theta v}$ *and* $[H_\theta, B_\theta] = 0$*. For each eigenpair* $H_\theta v = \beta v$*, the Fourier function satisfies*

$$(p_\theta U')' + (\alpha \cos x - \beta)U = 0, \qquad U(x + 2\pi) = \sigma U(x), \quad \sigma = (-1)^{N-1}. \tag{5}$$

The eigenvalues of H_θ *are simple, so its eigenvectors are also eigenvectors of* B_θ*.*

Proof. For a Fourier monomial e^{imx}, the coefficients at frequencies $m+1, m-1$ are respectively $[\alpha - m(m+1)]/2$ and $[\alpha - m(m-1)]/2$. The coefficient at m is $\cos\theta\, m^2$. These are precisely (2); the two outer coefficients vanish at the ends of the finite frequency set. Integration by parts on $[-\theta, \theta]$ gives $H_\theta B_\theta = B_\theta H_\theta$, since p_θ vanishes at both endpoints and the Fourier functions and their derivatives are bounded. The positive off-diagonal entries make H_θ an irreducible Jacobi matrix, with simple spectrum; $N = 1$ is immediate. Finally $e^{2\pi i m_j} = \sigma$. □

2 A normalized connection coefficient and its derivative

For a complex spectral parameter b, let $u_\theta(x; b)$ be the solution of

$$(p_\theta u')' + (\alpha \cos x - b)u = 0 \tag{6}$$

analytic at $x = -\theta$, with $u_\theta(-\theta; b) = 1$. At either endpoint the indicial root is zero with multiplicity two. The analytic solution with specified endpoint value is unique: its Taylor recurrence divides the coefficient of degree k by $k^2 \sin\theta \neq 0$. In particular, an analytic solution that vanishes at an endpoint vanishes identically.

Set $v_\theta(x; b) = u_\theta(-x; b)$, the analytic solution normalized to one at the right endpoint. Define $C_\theta(b)$ by the local form

$$u_\theta(x; b) = C_\theta(b) v_\theta(x; b) \log(\theta - x) + a_\theta(x; b), \tag{7}$$

where a_θ is analytic near θ. This fixes the logarithmic normalization. All derivatives of C_θ below are with respect to its own argument b, holding α and θ fixed.

Lemma 2 (Wronskian and norming identities)**.** *The function* C_θ *is entire in* b*, and*

$$C_\theta(b) = -\frac{2(1 - \cos\theta)}{\sin\theta} u_\theta(0; b)\, \partial_x u_\theta(0; b). \tag{8}$$

At a real zero b_0 *of* C_θ*, write* $h = u_\theta(\theta; b_0)$*. Then* $h \in \{-1, 1\}$ *and*

$$M_\theta(b_0) := \int_{-\theta}^{\theta} u_\theta(x; b_0)^2 \, \mathrm{d}x = -\sin\theta\, h\, C'_\theta(b_0) > 0. \tag{9}$$

In particular, every real zero of C_θ *is simple.*

Proof. The Wronskian $p_\theta(uv' - u'v)$ is constant. Its right-endpoint limit is $\sin\theta\, C_\theta$, while its value at zero is $-2(1 - \cos\theta) u_\theta(0) \partial_x u_\theta(0)$. This proves (8). The endpoint series is analytic in b; continuation across compact subintervals preserves that dependence. Thus both factors in (8), and hence C_θ, are entire.

At a real zero the solution is real and analytic at both endpoints, and $h \neq 0$ by endpoint uniqueness. Reflection and uniqueness give $u_\theta(-x; b_0) = h\, u_\theta(x; b_0)$; evaluation at the other endpoint gives $h^2 = 1$. Let $w = \partial_b u_\theta|_{b_0}$. Differentiating (6) and subtracting the two equations yields

$$\big[p_\theta(uw' - u'w)\big]' = u^2.$$

The left endpoint contributes zero. At the right endpoint, $w = C'_\theta(b_0) v_\theta \log(\theta - x) + O(1)$, and the boundary term is $-\sin\theta\, h\, C'_\theta(b_0)$. Integration proves (9), including its sign and nonvanishing. □

3 Exact concentration formula at every filling

Theorem 3 (Complementary-band connection formula)**.** *For every eigenmode $H_\theta v = \beta v$, $B_\theta v = \zeta v$, set $\bar\theta = \pi - \theta$. Then*

$$\boxed{\frac{\zeta}{1-\zeta} = (-1)^{N-1}\frac{C'_\theta(\beta)}{C'_{\bar\theta}(-\beta)}.} \tag{10}$$

Both denominator and numerator are nonzero. Equivalently,

$$\mathcal{P} = \log|C'_\theta(\beta)| - \log|C'_{\bar\theta}(-\beta)|. \tag{11}$$

Proof. The entire Fourier solution of (5) is nonzero at both $-\theta$ and θ; otherwise endpoint uniqueness would make it identically zero. Therefore $u_\theta(x;\beta) = U(x)/U(-\theta)$ is the real normalized regular solution, $C_\theta(\beta) = 0$, and $U(\theta) = hU(-\theta)$ with $h = \pm 1$.

Translate the complementary arc by $x = \pi + y$. Since $p_\theta(\pi + y) = -p_{\bar\theta}(y)$, the translated equation is (6) with (θ, b) replaced by $(\bar\theta, -\beta)$. Its near-normalized solution is $U(\pi + y)/U(\theta)$, and its far value is

$$\bar h = \frac{U(2\pi - \theta)}{U(\theta)} = \frac{\sigma}{h} = \sigma h.$$

The two arc normalizations have equal absolute value. Equation (3) therefore gives $\zeta/(1-\zeta) = M_\theta(\beta)/M_{\bar\theta}(-\beta)$. Apply (9), using $\sin\bar\theta = \sin\theta$ and $h/\bar h = \sigma$, to obtain (10). □

At half filling $\theta = \pi/2$, translating $s = x + \pi/2$ gives

$$(\sin s\, u')' + (\alpha \sin s - b)u = 0, \qquad 0 < s < \pi.$$

If J has bonds $j(N-j)/N$ and zero diagonal, then $H_{\pi/2} = NJ/2$ and $\beta = N\lambda/2$. For even N, (10) is precisely $\zeta/(1-\zeta) = -C'(\beta)/C'(-\beta)$. For odd N, the sign is positive. For example, at the central mode $\beta = 0$ one has $\zeta = 1/2$; a fixed minus sign would give -1 instead of 1. More explicitly, for $N = 2m+1$ the regular central solution is $P_m(\cos s)$ and

$$C'(0) = -(-1)^m \int_0^\pi P_m(\cos s)^2\, \mathrm{d}s \neq 0. \tag{12}$$

The norming integral here uses $\mathrm{d}s$, not $\sin s\, \mathrm{d}s$.

Corollary 4 (Removal of the finite characteristic polynomial)**.** *Let $D_\theta(b) = \det(bI - H_\theta)$ and $R_\theta(b) = C_\theta(b)/D_\theta(b)$, with removable values filled in. Then R_θ is entire, is nonzero at every eigenvalue of H_θ, and at each such eigenvalue*

$$\frac{\zeta}{1-\zeta} = \frac{R_\theta(\beta)}{R_{\bar\theta}(-\beta)} = (-1)^N \lim_{b\to\beta}\frac{C_\theta(b)}{C_{\bar\theta}(-b)}. \tag{13}$$

Proof. The N distinct zeros of D_θ are simple zeros of C_θ, so division removes only simple zeros. For $E_{jj} = (-1)^j$, $H_{\bar\theta} = -EH_\theta E$, whence $D_{\bar\theta}(-b) = (-1)^N D_\theta(b)$ and $D'_{\bar\theta}(-\beta) = \sigma D'_\theta(\beta)$. Substitute $C'_\theta(\beta) = D'_\theta(\beta)R_\theta(\beta)$ into (10). The last expression follows by taking the limit. □

This does not identify the full zero set of C_θ with the finite matrix spectrum. Nor is (13) asserted to be a concentration formula away from those spectral points.

4 Quantitative routes to a period remainder

The exact formula isolates the missing asymptotic task without solving it. Two elementary stability statements specify sufficient accuracy.

Theorem 5 (Relative-error criteria)**.** *At an exact eigenvalue* β*, put* $d_+ = C'_\theta(\beta)$*,* $d_- = C'_{\tilde\theta}(-\beta)$*. If nonzero approximations* $\hat d_\pm$ *satisfy* $|\hat d_\pm/d_\pm - 1| \le q_\pm < 1$*, then*

$$\left| \mathcal{P} - \log \left| \frac{\hat d_+}{\hat d_-} \right| \right| \le -\log(1-q_+) - \log(1-q_-). \tag{14}$$

Alternatively let $u_\pm$ *be the two near-normalized regular arc solutions in the theorem above. For approximants* $f_\pm$ *with* $F_\pm = \|f_\pm\|_2 > 0$*, suppose* $\|u_\pm - f_\pm\|_2 \le \delta_\pm F_\pm$*,* $0 \le \delta_\pm < 1$*. Writing* $\widehat{\mathcal{P}} = \log(F_+^2/F_-^2)$ *gives*

$$\begin{aligned} 2\log(1-\delta_+) - 2\log(1+\delta_-) &\le \mathcal{P} - \widehat{\mathcal{P}} \\ &\le 2\log(1+\delta_+) - 2\log(1-\delta_-). \end{aligned} \tag{15}$$

Proof. The reverse triangle inequality bounds $|\hat d_\pm/d_\pm|$ between $1-q_\pm$ and $1+q_\pm$. Apply (11) and $\log(1+q) \le -\log(1-q)$. For the second assertion, $(1-\delta_\pm)F_\pm \le \|u_\pm\|_2 \le (1+\delta_\pm)F_\pm$. Square, take the ratio, and use the norm formula in the proof of (10). □

These bounds do not contain a factor $1/[\zeta(1-\zeta)]$: relative norming data control the logit directly even near concentration endpoints. But the hypotheses require the correct endpoint normalization and the actual spectral parameter. A small residual for a homogeneous ODE alone cannot set the normalization: multiplying a solution by a constant preserves that equation and changes its squared norm. Use of an approximate β also needs an independent eigenvalue and parameter-sensitivity bound.

For the half-filled model with even N, set $t = 2\beta/N^2$ and define the finite formal-period truncation

$$P_N^{[5]}(t) = N\eta(t) - \frac{\varphi_1(t)}{N} - \frac{\varphi_3(t)}{N^3} - \frac{\varphi_5(t)}{N^5}. \tag{16}$$

Here $\eta(t) = \pi t\, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2)$ and the φ_k are the explicitly defined formal cycle coefficients in [3]. For a compact $K \subset (-1/2, 0) \cup (0, 1/2)$, sufficient conditions for the desired spectral theorem are uniform $q_\pm = O_K(N^{-7})$ in (14) and

$$\log|\hat d_+/\hat d_-| = P_N^{[5]}(t) + O_K(N^{-7})$$

at the modes with $t \in K$. Equation (15) provides an alternative through normalized solution norms. Either set of conditions implies

$$\left| e - \frac{\varphi_1(t)}{N} - \frac{\varphi_3(t)}{N^3} - \frac{\varphi_5(t)}{N^5} \right| \le C_K N^{-7}, \qquad e = -\mathcal{P} + N\eta(t). \tag{17}$$

The implication is proved; the stated uniform approximation hypotheses have not been established here. A formal all-orders series also needs a defined summation before it can be called an exact period function.

5 Verification and scope

The identities above have analytic proofs independent of the numerical checks. A separate computation obtains β and ζ from the finite matrices and computes C and C' from a parameter-differentiated Frobenius recurrence and the midpoint Wronskian (8). The concentration eigenvalue is not used in that connection calculation. At 90 decimal digits and Taylor degree 380, 74 cases at $\theta/\pi = 1/4, 1/2, 3/4$, with selected sizes $1 \le N \le 25$, gave a largest relative odds discrepancy below 3.43×10^{-73}. These comprised 48 half-filled cases and 26 other-filling cases. Four repetitions at 120 digits and degree 500 gave discrepancies below 9.77×10^{-114}; separate complex-step checks tested the parameter derivative. Exact low-order solutions also fix the logarithmic normalization. Ordinary multiprecision agreement is not interval validation.

These results refine the finite reference problem in *From Prephysical Hypotheses to Physical Models*. They do not establish the uniform period expansion, the full Wilson relative-entropy law, or a Yang–Mills mass-gap theorem. The exact complementary-band formula and the sufficient error criteria are the established conclusions here. The whole-line construction in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves the half-filled fixed-interior expansion. For interval restrictions of the specified infinite-chain Wilson–Dirac projectors with $m = \mu/L$ and fixed $\mu > 0$, Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* determines the finite part with remainder $O_\mu(1/\log L)$. Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* also proves $\|E_N\| \le 7$ for even $N \ge 2$. These conclusions do not establish the sufficient endpoint approximation hypotheses stated in this article, a general-filling expansion, or a moving-edge expansion.

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Spectral Tails and Exact Concentration Logits

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Abstract

For the half-filled finite sine kernel, we express the concentration logit as a convergent sum over the spectrum of a singular differential operator after its first N eigenvalues have been removed. The finite Fourier subspace is proved to exhaust those first N modes. A Green-kernel trace identity then determines the logarithmic derivative of the deflated connection coefficient, with no undetermined entire factor. The resulting matrix identity is an absolutely convergent odd-power series with positive coefficients. It gives exact sign and monotonicity statements and explicit spectral-tail truncation bounds at fixed N. Formal center coefficients are compared with this representation. Their large-N interpretation is developed in the reciprocal-trace companions, and the fixed-interior quantum-period remainder is proved separately in *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*.

1 Finite reference problem

Fix an integer $N \geq 1$. For $0 \leq j, k < N$, let

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad H_{j-1,j} = H_{j,j-1} = \frac{j(N-j)}{2}, \quad H_{jj} = 0. \tag{1}$$

This is the classical discrete concentration geometry used in the free-fermion entanglement problem [1]. We use

$$\mathcal{P}_N = \log B - \log(I - B), \qquad K_N = -\mathcal{P}_N, \tag{2}$$

with natural logarithms. The kernel is strictly between zero and one: its quadratic form and that of its complement are integrals of the squared absolute value of a nonzero finite Fourier polynomial over nonempty arcs. No physical clock or energy scale is assumed.

Put $a = \pi/2$, $I = (-a, a)$, $\alpha = (N^2 - 1)/4$, and

$$U_v(x) = \sum_{j=0}^{N-1} v_j e^{i[j-(N-1)/2]x}. \tag{3}$$

Direct Fourier summation gives

$$\mathcal{L}_N U_v = U_{Hv}, \qquad \mathcal{L}_N u = (\cos x\, u')' + \alpha \cos x\, u. \tag{4}$$

The outer Fourier coefficients vanish because $\alpha - m(m \pm 1) = 0$ at the appropriate endpoints of the frequency set. Integration by parts on I, where $\cos x$ vanishes at the endpoints, proves $HB = BH$. Since H has positive bonds, it has simple spectrum. Its real eigenvectors therefore give common eigenpairs $Hv = \beta v$, $Bv = \zeta v$, with $0 < \zeta < 1$.

The diagonal matrix $E_{jj} = (-1)^j$ obeys

$$EHE = -H, \qquad EBE = I - B. \tag{5}$$

Thus the spectrum of H is symmetric. Write $h_N = \|H\|$. The case $N = 1$ has $h_N = 0$ and $\mathcal{P}_N = 0$.

2 The differential spectrum and its first finite block

Let A_N be the self-adjoint Friedrichs operator associated with the closure on $C_c^\infty(I)$ of

$$\mathfrak{a}_N[f] = \int_I \cos x\, |f'|^2 \,\mathrm{d}x - \alpha \int_I \cos x\, |f|^2 \,\mathrm{d}x. \tag{6}$$

The differential expression is $A_N = -\mathcal{L}_N$. Its eigenfunctions use the regular, logarithm-free solution at both endpoints. We justify this choice below rather than selecting a different self-adjoint extension implicitly. The role of this condition in singular Sturm–Liouville problems is classical [2, 3].

Lemma 1 (Compact spectrum and elementary bounds)**.** *The spectrum of A_N is simple and discrete. In increasing order, with zero-based indices, write it as $\mu_0 < \mu_1 < \cdots$. For $c = 2/\pi^2$,*

$$c\,k(k+1) - \alpha \;\le\; \mu_k \;\le\; \frac{1}{2}k(k+1). \tag{7}$$

In particular its resolvent is trace class away from the spectrum.

Proof. For $|x| < a$,

$$\frac{\cos x}{a^2 - x^2} = \frac{1}{2}\frac{\sin[(a-x)/2]}{(a-x)/2}\frac{\sin[(a+x)/2]}{(a+x)/2}$$

lies between $2/\pi^2$ and $1/2$. Thus the positive derivative form has the same form domain and equivalent form norm as that of the rescaled Friedrichs Legendre operator $-[(a^2 - x^2)f']'$, whose eigenvalues are $k(k+1)$ [3]. That form domain embeds compactly into $L^2(I)$. The potential $-\alpha\cos x$ is bounded between $-\alpha$ and zero. The min–max principle now gives compactness and (7).

At either endpoint, the two local solution types of the eigen-equation are analytic and analytic times a logarithm plus analytic. The logarithmic type has infinite derivative energy $\int_0^\epsilon r\,|1/r|^2\,\mathrm{d}r$ and is excluded by the form domain. The regular type belongs to the closure: a cutoff from zero to one over $\epsilon^2 < r < \epsilon$, linear in $\log r$, has derivative energy $O(1/|\log\epsilon|)$ and vanishing L^2 error. For these regular solutions the endpoint flux vanishes, so the eigen-equation also holds in the operator domain. The regular solution at the left endpoint is unique up to a constant; hence each eigenvalue is simple. This argument concerns eigenfunctions, not arbitrary form-domain functions, which need not be bounded at the endpoints. Finally (7) implies summability of the resolvent eigenvalues. □

Theorem 2 (Exact selection of the first N modes)**.** *The eigenvalues of A_N on its first N modes are exactly the negatives of the eigenvalues of H. Consequently*

$$\mu_{N-1} = h_N, \qquad \mu_N > h_N \ge 0. \tag{8}$$

Proof. Let V_N be the real-valued part of the centered Fourier space spanned by the functions in (4). Its real dimension is N; it is invariant under A_N, and its eigenfunctions are regular at both endpoints. Thus it supplies N distinct eigenvalues.

A real continuous function orthogonal to V_N must have at least N sign changes if it has only finitely many zeros. Indeed, if its sign-change points are $z_1, \dots, z_r$ with $r < N$, then

$$q(x) = \prod_{\ell=1}^{r} \sin\frac{x - z_\ell}{2}\,\cos^{N-1-r}\frac{x}{2}$$

belongs to V_N. On I, it has exactly those sign changes. The product of the function and q has a fixed sign and nonzero integral, contradicting orthogonality.

The eigenfunction of index k has at most $k+1$ nodal domains. For completeness, its restrictions to nodal domains lie in the closed form domain and have disjoint supports, each with Rayleigh quotient μ_k. If there were more than $k+1$ such pieces, their span would contain a nonzero vector orthogonal to the first $k+1$ eigenfunctions. Its quotient would be both μ_k and at least μ_{k+1}, a

contradiction. Interior zeros are simple, and endpoint regularity prevents an accumulation there, so the sign-change argument applies.

An eigenfunction of index $k < N$ outside V_N would be orthogonal to the invariant finite space, and would have at least N sign changes, contradicting the nodal bound. Hence V_N contains the first N modes. Symmetry (5) identifies the largest selected value as h_N, and simplicity gives the strict inequality in (8). □

3 Connection coefficients and a resolvent trace

Let $u(x;b)$ solve

$$(\cos x\, u')' + (\alpha \cos x - b)u = 0, \qquad u(-a;b) = 1, \tag{9}$$

with the regular left endpoint condition. Let $v(x;b) = u(-x;b)$. At the right endpoint define $C_N(b)$ by

$$u(x;b) = C_N(b)v(x;b)\log(a-x) + \text{an analytic function}.$$

The regular right factor has value one at a. The normalized Frobenius recursion and continuation show that C_N is entire in b, and its constant Wronskian is

$$W = \cos x\,(uv' - u'v) = C_N(b). \tag{10}$$

In particular $C_N(b) = -2u(0;b)u'(0;b)$. Its zeros are precisely $b = -\mu_k$. They are simple: at a real zero, reflection gives $u(a) = h = \pm 1$, and parameter differentiation yields

$$\int_I u^2 \,\mathrm{d}x = -h\, C_N'(b) > 0. \tag{11}$$

There are no nonreal zeros, because a regular solution at both endpoints would give an eigenvalue of the self-adjoint operator.

Lemma 3 (Trace identity without a normalization ambiguity)**.** *For b outside the zeros of C_N,*

$$\frac{C_N'(b)}{C_N(b)} = \mathrm{Tr}(A_N + b)^{-1} = \sum_{k=0}^{\infty} \frac{1}{\mu_k + b}. \tag{12}$$

Proof. The Green kernel of $A_N + b$ is $G_b(x,y) = -u(\min(x,y);b)v(\max(x,y);b)/C_N(b)$. The minus sign makes the derivative jump of $-\cos x\, \partial_x G$ equal to one. Differentiating (9) with respect to b and pairing with v gives

$$\left[\cos x\,(u_b v' - u_b' v)\right]' = -uv.$$

The boundary values are zero at $-a$ and $C_N'(b)$ at a, so $C_N'(b) = -\int_I uv\,\mathrm{d}x$.

For real $b > \alpha$, the resolvent is positive and trace class. On closed interior subintervals its continuous positive kernel has trace equal to the integral of its diagonal. Exhausting I by those subintervals and using trace convergence for a positive trace-class operator gives $\mathrm{Tr}(A_N + b)^{-1} = \int_I G_b(x,x)\,\mathrm{d}x$. This proves the first equality there. Both sides extend meromorphically, so the equality holds throughout their common domain. The series equality is the spectral theorem with (7). □

Let $D_N(b) = \det(bI - H)$ and $R_N(b) = C_N(b)/D_N(b)$, with its removable values filled in. The first-N theorem and (12) give the exact cancellation

$$\frac{R_N'(b)}{R_N(b)} = \sum_{k=N}^{\infty} \frac{1}{\mu_k + b}. \tag{13}$$

In particular R_N is nonzero on $(-\mu_N, \mu_N)$. There is no unspecified exponential factor: (12) fixes the logarithmic derivative directly. Related Wronskian determinant methods are classical [4]; no regularized determinant is needed in this argument.

4 Exact positive expansion and truncation bounds

Theorem 4 (Spectral-tail formula)**.** *Define, for* $|b| < \mu_N$*,*

$$F_N(b) = 2\sum_{k=N}^{\infty} \operatorname{artanh}\frac{b}{\mu_k} = \sum_{k=N}^{\infty} \log\frac{\mu_k + b}{\mu_k - b}. \tag{14}$$

The series converges locally uniformly. At every common finite eigenmode,

$$\log\frac{\zeta}{1-\zeta} = F_N(\beta). \tag{15}$$

Consequently the following series converges absolutely in matrix operator norm:

$$\boxed{\mathcal{P}_N = 2\sum_{k=N}^{\infty} \operatorname{artanh}(H/\mu_k)} = 2\sum_{r=0}^{\infty} \frac{Z_N(2r+1)}{2r+1} H^{2r+1}, \tag{16}$$

$$Z_N(s) = \sum_{k=N}^{\infty} \mu_k^{-s}, \qquad s \geq 1.$$

Every coefficient on the right is strictly positive.

Proof. We first recover the finite concentration ratio from the normalized connections. At a selected root $b = \beta$, the Fourier solution divided by its nonzero left endpoint equals $u(x;\beta)$ and has right endpoint $h = \pm 1$. On the complementary arc, translation by π gives the same equation with parameter $-\beta$. The centered Fourier function satisfies $U(x+2\pi) = \sigma U(x)$, where $\sigma = (-1)^{N-1}$. Thus the complementary normalized endpoint value is σh. Parseval and (11) imply

$$\frac{\zeta}{1-\zeta} = \sigma\frac{C_N'(\beta)}{C_N'(-\beta)} = \frac{R_N(\beta)}{R_N(-\beta)}.$$

The last equality uses $D_N(-b) = (-1)^N D_N(b)$ and its derivative; it also covers the central zero for odd N by removable values.

Integrating (13) from $-b$ to b gives (14) as $\log|R_N(b)/R_N(-b)|$. The zero-free interval fixes the positive ratio, and $\sum_{k\geq N} 1/\mu_k < \infty$ justifies integration term by term. This proves (15). Because $h_N/\mu_N < 1$, the scalar artanh expansions converge uniformly on the whole finite spectrum. Their absolute norm sums are bounded by $2\sum_{k\geq N} \operatorname{artanh}(h_N/\mu_k) < \infty$. Functional calculus and absolute convergence prove (16). □

The function F_N is odd and strictly increasing. For $0 < b < \mu_N$, $F_N(b)/b$ is strictly increasing, by its positive Taylor coefficients. In particular $\operatorname{sgn}(2B - I) = \operatorname{sgn} H$, including the zero mode. For $N \geq 2$, (16) also gives zero entries between sites of equal parity and strictly positive entries between sites of opposite parity: every odd power of the positive-bond matrix H is entrywise nonnegative, and an odd-length path connects every opposite-parity pair. The signs reverse for K_N. These are finite-N conclusions. The leading scaling-limit power series in [1] is a related result with different coefficients.

Corollary 5 (Explicit omitted-spectrum bound)**.** *Let* $0 \leq b \leq h_N$ *and let* $M \geq N$ *be the first omitted index. If* $\rho = (\alpha + b)/[cM(M+1)] < 1$*, then*

$$\frac{4b}{M} \;\leq\; F_N(b) - 2\sum_{k=N}^{M-1} \operatorname{artanh}(b/\mu_k) \;\leq\; \frac{2b}{c(1-\rho)M}. \tag{17}$$

For negative b*, oddness gives the corresponding signed bounds. The upper bound with* $b = h_N$ *also bounds the matrix operator-norm truncation error in* (16)*.*

Proof. Use $z \leq \operatorname{artanh} z \leq z/(1-z)$ for $0 \leq z < 1$. The upper eigenvalue bound in (7) gives the lower estimate. For the upper estimate, $\mu_k - b \geq c(1-\rho)k(k+1)$ for $k \geq M$. Now $\sum_{k=M}^{\infty} 1/[k(k+1)] = 1/M$. The norm statement follows from the spectral theorem and monotonicity. □

The bound concerns the exact differential eigenvalues. Computed approximations require their own error enclosures; an ordinary Galerkin calculation does not automatically make this a numerical certificate.

Corollary 6 (Acceleration and a quantitative certificate)**.** *For* $0 \le b \le h_N$*, set* $g_b(\mu) = 2[\operatorname{artanh}(b/\mu) - b/\mu]$*. Then*

$$F_N(b) = 2bZ_N(1) + \sum_{k=N}^{\infty} g_b(\mu_k). \tag{18}$$

Let $M \ge N$*,* $d = c - \alpha/[M(M+1)] > 0$*, and* $q = b/[dM(M+1)] < 1$*. The omitted sum obeys*

$$0 \le \sum_{k=M}^{\infty} g_b(\mu_k) \le \frac{2b^3}{15(1-q^2)d^3M^5}. \tag{19}$$

If $|\widehat{Z} - Z_N(1)| \le \delta_Z$ *and* $|\hat{\mu}_k - \mu_k| \le \delta_k$ *for* $N \le k < M$*, with* $a_k = \hat{\mu}_k - \delta_k > b$*, the absolute error of the resulting finite approximation is at most*

$$2b\delta_Z + \sum_{k=N}^{M-1} \frac{2b^3\delta_k}{a_k^2(a_k^2 - b^2)} + \frac{2b^3}{15(1-q^2)d^3M^5}. \tag{20}$$

Proof. For $0 \le z \le q < 1$, the positive series gives $\operatorname{artanh} z - z \le z^3/[3(1-q^2)]$. For $k \ge M$, (7) gives $\mu_k \ge d\,k(k+1)$. Moreover

$$\frac{1}{[k(k+1)]^3} \le \frac{1}{5}\left(\frac{1}{k^5} - \frac{1}{(k+1)^5}\right),$$

since $(k+1)^5 - k^5 - 5k^2(k+1)^2 = 5k^2 + 5k + 1 > 0$. Summing proves (19). Finally $|g_b'(\mu)| = 2b^3/[\mu^2(\mu^2 - b^2)]$ decreases for $\mu > b$. The mean-value theorem and the triangle inequality give (20). □

5 Center law: exact structure and formal predictions

Let $J = 2H/N$, $Jv = \lambda v$, and $t = \lambda/N = 2\beta/N^2$. Write

$$\varepsilon = \log\frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(t), \qquad \eta(t) = \pi t\,{}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2).$$

The exact involution sends (t, ε, e) to $(-t, -\varepsilon, -e)$. Thus a leading relation involving $e/|\varepsilon|$ cannot have the same positive sign on both branches. Symmetry alone does not fix the sign or magnitude of e/ε.

The previously reduced formal coefficients [5] give

$$\begin{aligned} \eta(t) &= \pi t + \tfrac{\pi}{4}t^3 + O(t^5), & \varphi_1(t) &= \tfrac{\pi}{4}t + \tfrac{25\pi}{32}t^3 + O(t^5), \\ \varphi_3'(0) &= -\tfrac{17\pi}{64}, & \varphi_5'(0) &= \tfrac{169\pi}{256}. \end{aligned} \tag{21}$$

Therefore the proposed expansion $\varepsilon = -N\eta + \varphi_1/N + \varphi_3/N^3 + \cdots$ has the sign-consistent formal center law

$$e \sim \frac{\pi\lambda}{4N^2} \sim -\frac{\varepsilon}{4N^2}. \tag{22}$$

This coefficient calculation does not prove a plunge asymptotic: a fixed-interior expansion away from $t = 0$ cannot simply be evaluated along a sequence of modes tending to zero.

The tail representation gives an exact, distinguished interpolation with

$$F_N'(0) = 2Z_N(1). \tag{23}$$

Differentiating the *formal candidate* at zero consequently suggests the separate trace asymptotic

$$Z_N(1) \overset{\text{formal}}{\sim} \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \cdots. \tag{24}$$

Neither the spectral period conjecture at discrete points nor a value-only remainder estimate alone justifies differentiating an interpolant. Equation (24) is an additional formal prediction to be tested and proved independently. For even N, it can be tested directly through $Z_N(1) = C_N'(0)/C_N(0)$, since D_N is even and nonzero at zero. For odd N, the removable-value formula is $Z_N(1) = C_N''(0)/(2C_N'(0))$.

One further exact formal coefficient. The linear center coefficient can be continued one order without assuming a spectral asymptotic. Set $p = \cos x$, $q = (p - 2t)/(4p)$, $r = 1/(4p^2)$, and solve the formal equation

$$W^2 + \tfrac{1}{2}(\log W)'' - \tfrac{1}{4}[(\log W)']^2 = N^2 q + r, \quad W = N\sqrt{q}\,(1 + a_1 N^{-2} + a_2 N^{-4} + a_3 N^{-6} + a_4 N^{-8} + \cdots). \tag{25}$$

Primes in (25) mean x derivatives. Coefficient comparison in the ring $\mathbb{Q}(p)[t]/(t^2)$, using $\partial_x^2 f = (1-p^2) f_{pp} - p f_p$, gives

$$\begin{aligned} a_4(p,0) &= \frac{4096p^6 - 143616p^4 + 493824p^2 - 375733}{128p^8}, \\ \partial_t a_4(p,0) &= -\frac{16p^8 - 50952p^6 + 825534p^4 - 2268425p^2 + 1553721}{16p^9}. \end{aligned} \tag{26}$$

The recursion fixes each new coefficient linearly, so these expressions are checked by substitution after the three preceding orders. For the counterclockwise period convention $\varphi_7 = i \oint \sqrt{q}\, a_4 \, \mathrm{d}x$ with $\mathrm{d}x = -\mathrm{d}p/\sqrt{1-p^2}$, a fixed circle $2|t| < R < 1$ yields

$$\varphi_7'(0) = \pi \operatorname*{Res}_{p=0} \frac{\partial_t a_4(p,0) - a_4(p,0)/p}{\sqrt{1-p^2}} = -\frac{50777\pi}{16384}. \tag{27}$$

This is an exact formal coefficient identity. The associated N^{-9} coefficient suggested for (24) is $50777\pi/16384 = 9.7363678083\ldots$; that spectral asymptotic is not proved by the residue calculation.

6 Scope and verification

Direct finite-matrix center-mode calculations at $N = 32, 64, 128, 256, 512, 1024$ give $-4N^2 e/\varepsilon$ increasing from 0.9993955762 to 0.9999992990 at 100-digit working precision. An independent connection calculation of $Z_N(1)$ at $N = 8, 16, 32, 64, 128$ gives the residual after the four displayed terms in (24), multiplied by N^9, as

$$8.7298271, \quad 9.4540545, \quad 9.6634554, \quad 9.7179853, \quad 9.7317624.$$

These values are consistent with the coefficient in (27). At $N = 128$, increasing precision from 130 to 180 digits and the Taylor degree from 600 to 800 changes the retained trace by less than 3.88×10^{-90}. These are numerical checks, not error enclosures or proofs of the limits.

The proofs above identify an exact finite reference function and give convergent spectral representations. They do not establish the large-N expansion of the remaining differential spectrum, the uniform quantum-period remainder, the full Wilson relative-entropy law, or a field-theory mass gap. The decomposition isolates a further precise target: estimate the positive traces $Z_N(2r+1)$ with uniform control sufficient to compare (16) to the declared period candidate. The reciprocal-trace, positive-moment and whole-line results supply the stated fixed-interior estimates. For the specified infinite-chain Wilson comparison at fixed positive scaled mass, Theorem 1 of *Finite Part of Wilson–Dirac Relative Entropy* gives the finite part and an $O_\mu(1/\log L)$ remainder; Theorem 3 of *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals* supplies the even-dimensional bound $\|E_N\| \le 7$. A moving-edge expansion remains open. Neither the finite-part theorem nor that norm bound promotes the formal coefficient calculation above to an additional spectral asymptotic. Separate reproduction material records numerical discretizations, precision comparisons, formal coefficient checks and their limitations.

Acknowledgments and provenance. This work develops the finite reference problem in *From Prephysical Hypotheses to Physical Models*. The earlier half-filled connection proposal was supplied in correspondence prepared with Claude (Anthropic). The present spectral-tail argument and checks were developed with ChatGPT (OpenAI). The wider program has also used Gemini (Google), Grok (SpaceXAI), and DeepSeek (High-Flyer); no particular proof here is attributed to those systems. Assistance implies neither external peer review nor endorsement. No priority claim is made.

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Reciprocal Traces and the Half-Filled Plunge Law

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Abstract

For the half-filled finite sine kernel at even size, the first reciprocal trace of the deflated differential spectrum is an exact positive Laplace transform. A weighted Legendre resolvent and a Mellin contour identity determine its normalization. Taylor estimates on a fixed analytic kernel then prove its large-size expansion to every fixed order, including an explicit remainder after the term of order N^{-9}. This result yields the signed plunge law for every nonzero Jacobi eigenmode whose eigenvalue tends to zero. An elementary trial vector proves that the central pair has this property. No uniform bulk quantum-period expansion is assumed.

1 The finite problem and its exact differential trace

Let $N \geq 2$ be even. Define real symmetric matrices, with indices $0 \leq j, k < N$, by

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad H_{j-1,j} = H_{j,j-1} = \frac{j(N-j)}{2}, \quad H_{jj} = 0, \qquad J_N = \frac{2H}{N}.$$

These are the usual half-filled concentration matrix and its commuting Jacobi matrix [2]. They commute, $0 < B < I$, and their common eigenpairs are denoted

$$Hv = \beta v, \qquad J_N v = \lambda v, \qquad Bv = \zeta v, \qquad \beta = N\lambda/2.$$

Put $\alpha = (N^2 - 1)/4$. On $(-\pi/2, \pi/2)$ consider the Friedrichs realization

$$A_N u = -(\cos x\, u')' - \alpha \cos x\, u.$$

Its eigenfunctions are regular, with no logarithmic term, at both endpoints. Write its increasing eigenvalues as $\mu_{N,k}$, starting at $k = 0$, and define

$$Z_N = \sum_{k=N}^{\infty} \mu_{N,k}^{-1}. \tag{1}$$

The exact spectral-tail construction [1] gives

$$\{\mu_{N,0}, \dots, \mu_{N,N-1}\} = -\operatorname{spec} H, \qquad \mu_{N,N} > \|H\|, \tag{2}$$

and the norm-convergent scalar representation at every common mode

$$\log \frac{\zeta}{1-\zeta} = F_N(\beta), \qquad F_N(b) = 2 \sum_{k=N}^{\infty} \operatorname{artanh} \frac{b}{\mu_{N,k}}, \quad |b| < \mu_{N,N}. \tag{3}$$

For clarity, the ingredients of these exact inputs are as follows. The centered finite Fourier space is invariant under $-A_N$ and supplies $-\operatorname{spec} H$. A sign-change test against that space, together with the variational nodal bound, identifies exactly the first N differential eigenvalues. For the left regular solution normalized to one, let $C_N(b)$ be its logarithmic coefficient at the right endpoint.

Its Green kernel gives $C_N'/C_N = \mathrm{Tr}(A_N + b)^{-1}$. Deflating $D_N(b) = \det(bI - H)$ removes precisely those first N terms. The endpoint Lagrange identity and the complementary Fourier arc give the occupation ratio as $[C_N/D_N](\beta)/[C_N/D_N](-\beta)$. Integrating the deflated resolvent trace proves (3). These statements concern the specified endpoint realization, not an arbitrary self-adjoint extension.

At $b = 0$ set $\nu = (N-1)/2$ and translate $s = x + \pi/2$. The regular solutions are $P_\nu(\cos s)$ and $P_\nu(-\cos s)$, where P_ν denotes the Ferrers function on $(-1, 1)$. The Legendre connection formula gives $C_N(0) = 2\sin(\pi\nu)/\pi$, while the parameter Lagrange identity gives $C_N'(0) = -\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s$. Because D_N is even and nonzero at zero,

$$Z_N = \frac{C_N'(0)}{C_N(0)} = -\frac{\pi}{2\sin\pi\nu}\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s. \tag{4}$$

The notation $P_{-1/2}(\cosh t)$ below instead denotes the real Legendre function for argument at least one. All spectral and modular quantities here are dimensionless; no physical clock or energy scale is introduced.

2 A weighted resolvent and a Mellin reflection identity

Define

$$\Phi(z) = {}_2F_1(1/2, 1/2; 1; z), \qquad G(z) = \Phi(z)^2, \qquad c_j = \frac{(1/2)_j}{j!}, \qquad a_l = \sum_{j=0}^{l} c_j^2 c_{l-j}^2.$$

Thus $G(z) = \sum_{l\geq 0} a_l z^l$ for $|z| < 1$.

Lemma 1 (Weighted Legendre trace)**.** *Let*

$$M(a) = \sum_{l=0}^{\infty} \frac{a_l}{l + a}.$$

This defines a meromorphic function with possible poles only at the nonpositive integers. For even N, with $\nu = (N-1)/2$,

$$Z_N = \frac{\pi}{2}\{M(-\nu) + M(\nu + 1)\}. \tag{5}$$

For $\Re a > 0$, $M(a) = \int_0^1 z^{a-1}G(z)\mathrm{d}z$.

Proof. The generating function for Legendre polynomials yields

$$P_l(\cos s) = \sum_{j=0}^{l} c_j c_{l-j} e^{i(2j-l)s}, \qquad \int_0^\pi P_l(\cos s)^2 \mathrm{d}s = \pi a_l.$$

Induction gives $c_j^2 \leq 1/(j+1)$, and hence

$$0 < a_l \leq \frac{2H_{l+1}}{l+2}, \qquad H_r = \sum_{j=1}^{r} \frac{1}{j}. \tag{6}$$

Consider the ordinary Legendre operator with regular/Friedrichs endpoint conditions, $L = -[(1 - x^2)u']'$ on $L^2(-1, 1; \mathrm{d}x)$, with eigenvalues $l(l+1)$ and normalized eigenfunctions $\sqrt{(2l+1)/2}P_l(x)$. Its shifted Green kernel has diagonal

$$(L - \nu(\nu+1))^{-1}(x, x) = -\frac{\pi}{2\sin\pi\nu}P_\nu(x)P_\nu(-x).$$

Initially $-1/2 < \nu < 0$, so the resolvent is positive. Integrating its diagonal with weight $(1 - x^2)^{-1/2}$ gives, by positive-kernel trace on compact intervals and monotone exhaustion,

$$-\frac{\pi}{2\sin\pi\nu}\int_0^\pi P_\nu(\cos s)P_\nu(-\cos s)\mathrm{d}s = \frac{\pi}{2}\sum_{l\geq 0} a_l\left(\frac{1}{l-\nu} + \frac{1}{l+\nu+1}\right).$$

The series is locally normally convergent away from its poles by (6). Endpoint logarithmic bounds, locally uniform in ν, give the meromorphic continuation of the integral expression. Continuation to the required half-integer ν and (4) prove (5). The Mellin integral for M follows by absolute convergence. At negative half-integers it always means meromorphic continuation, not a divergent ordinary integral. □

Lemma 2 (Mellin reflection at half-integers)**.** *For $n \geq 1$,*

$$M(1/2-n) + M(1/2+n) = \int_0^1 z^{n-1/2} G(1-z)\mathrm{d}z. \tag{7}$$

Proof. For $0 < a < 1$, introduce

$$J(c) = \int_0^1 z^{c-1} G(1-z)\mathrm{d}z, \qquad Q(a) = \int_0^\infty x^{a-1} G(-x)\mathrm{d}x.$$

Use the principal branch of Φ, cut along $[1,\infty)$. Reciprocal-modulus continuation [3] gives, for $x > 1$,

$$\Phi(x+i0) = x^{-1/2}\{\Phi(1/x) + i\Phi(1-1/x)\}, \quad \Re G(x+i0) = x^{-1}\{G(1/x) - G(1-1/x)\}.$$

Integrate $z^{a-1}G(z)$ around the upper half-plane, with small indentations at zero and one. The origin arc is $O(\epsilon^a)$; the arc at one is $O(\epsilon(1+|\log\epsilon|)^2)$. Since $\Phi(z) = O(|z|^{-1/2}(1+\log|z|))$ there, the large arc is $O(R^{a-1}(1+\log R)^2)$ and also vanishes. The negative-axis phase therefore gives

$$\int_0^\infty x^{a-1} G(x+i0)\mathrm{d}x = e^{i\pi a} Q(a).$$

Taking real parts and changing $z = 1/x$ on $x > 1$ yields

$$M(a) + M(1-a) - J(1-a) = \cos(\pi a) Q(a). \tag{8}$$

Pfaff's transformation gives $G(-x) = (1+x)^{-1} G(x/(1+x))$, so

$$Q(a) = \int_0^1 z^{a-1}(1-z)^{-a} G(z)\mathrm{d}z.$$

For $\Re a < 1$, its endpoint at one is integrable. At zero, subtracting finitely many Taylor terms of $(1-z)^{-a}G(z)$ continues this integral meromorphically; its possible poles in that half-plane are only $0, -1, -2, \ldots$. Both sides of (8) consequently continue to $\Re a < 1$. At $a = 1/2 - n$, $Q(a)$ is finite and $\cos(\pi a) = 0$. This proves (7), including its normalization. □

3 Exact positive representation and trace asymptotics

Theorem 3 (Positive Laplace formula)**.** *For every positive even integer N,*

$$\boxed{Z_N = \pi \int_0^\infty e^{-Nt} f(t)\mathrm{d}t, \qquad f(t) = P_{-1/2}(\cosh t)^2.} \tag{9}$$

Equivalently, $Z_N = (\pi/2)\int_0^1 z^{(N-1)/2} G(1-z)\mathrm{d}z$.

Proof. The equivalent form follows immediately from (5) and (7). The quadratic transformation

$$y(t) := P_{-1/2}(\cosh t) = e^{-t/2}\Phi(1-e^{-2t})$$

then gives (9) on substituting $z = e^{-2t}$. The transformation can be checked directly: both sides are the unique regular solution of

$$y'' + \coth t\, y' + \tfrac{1}{4} y = 0, \qquad y(0) = 1, \quad y'(0) = 0. \tag{10}$$

Thus no numerical match or asymptotic normalization is required. □

Euler's integral gives the particularly useful real formula

$$y(t) = \frac{2}{\pi}\int_0^{\pi/2} \frac{\mathrm{d}\varphi}{\sqrt{1+\sinh^2(t/2)\sin^2\varphi}}. \tag{11}$$

It follows that $0 < f(t) \le 1$, with strict decrease for $t > 0$. The right side of (9) defines a completely monotone interpolation for real $N > 0$, but its identification with the spectral trace is asserted here only at even integers. In particular, odd N has a zero finite mode and requires a different deflation at $b = 0$.

Corollary 4 (An exact slope bound)**.** *For even $N \ge 2$,*

$$0 < \pi - NZ_N < \frac{\pi}{4N^2}. \tag{12}$$

Moreover NZ_N/π increases strictly along the positive even integers and tends to one.

Proof. Equation (10) implies

$$-y'(t) = \frac{1}{4\sinh t}\int_0^t \sinh s\, y(s)\mathrm{d}s < \tfrac{1}{4}\tanh(t/2) < t/8 \quad (t > 0).$$

Thus $0 < 1-f(t) = (1-y(t))(1+y(t)) < t^2/8$. Integration proves (12). Finally $NZ_N/\pi = \int_0^\infty e^{-u} f(u/N)\mathrm{d}u$; strict monotonicity and dominated convergence give the remaining claims. □

Theorem 5 (Asymptotic expansion with a remainder)**.** *As positive even $N \to \infty$,*

$$\boxed{Z_N = \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \frac{50777\pi}{16384N^9} + O(N^{-11}).} \tag{13}$$

There is an ordinary asymptotic expansion to every fixed order. In (13), the absolute remainder is at most $\pi B\, 10!/N^{11}$, where

$$\rho = 3/2, \qquad M = (1-\sinh^2(3/4))^{-1},$$
$$B = \max\left\{\frac{M\rho^{-10}}{1-\rho^{-2}},\ 2 + \frac{1}{8} + \frac{17}{1536} + \frac{169}{184320} + \frac{50777}{660602880}\right\}.$$

Proof. Expansion of (10), or of its defining hypergeometric series, gives

$$f(t) = 1 - \frac{t^2}{8} + \frac{17t^4}{1536} - \frac{169t^6}{184320} + \frac{50777t^8}{660602880} + O(t^{10}). \tag{14}$$

For complex $|t| \le \rho$, $|\sinh(t/2)| \le \sinh(3/4) < 1$. Equation (11), with the square root continued from zero, shows that f is even and analytic on a neighborhood of this disk, with $|f| \le M$. Cauchy's estimate therefore bounds its coefficient of t^{2j} by $M\rho^{-2j}$. Let S_8 be the polynomial displayed in (14). For $0 \le t \le 1$, the geometric coefficient tail gives $|f(t) - S_8(t)| \le M\rho^{-10}t^{10}/(1-\rho^{-2})$. For $t \ge 1$, $0 < f \le 1$ gives the other term defining B. Thus $|f - S_8| \le Bt^{10}$ on the entire positive axis. Integrating this finite Taylor estimate in (9) proves the stated remainder and coefficients. The same argument applies at each fixed truncation order. It does not assert convergence of the infinite inverse-power series. □

The coefficient of N^{-9} is now a spectral-trace theorem. Its agreement with a formal quantum-period coefficient does not, by itself, prove that period expansion at other spectral modes.

4 A uniform estimate near the center

Define the leading reference function and errors by

$$\eta(t) = \pi t\, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \qquad \varepsilon = \log\frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(\lambda/N). \tag{15}$$

Theorem 6 (Signed plunge law for shrinking Jacobi modes)**.** *Let even $N \to \infty$ and choose nonzero common eigenmodes with $\lambda_N \to 0$. Then*

$$e_N = \frac{\pi\lambda_N}{4N^2}\left[1 + O(N^{-2} + \lambda_N^2)\right], \qquad \varepsilon_N = -\pi\lambda_N\left[1 + O(N^{-2})\right]. \tag{16}$$

Consequently

$$\boxed{\frac{-4N^2 e_N}{\varepsilon_N} \longrightarrow 1.} \tag{17}$$

The estimates apply to either sign of the eigenvalue.

Proof. We give a bound uniform for even $N \geq 8$ and $0 < |\lambda| \leq 1$. The central two-site Rayleigh quotient gives $\|H\| \geq N^2/8$. By (2),

$$r := \frac{|\beta|}{\mu_{N,N}} \leq \frac{4|\lambda|}{N} \leq \frac{1}{2}, \qquad \sum_{k\geq N} \mu_{N,k}^{-3} \leq \frac{64 Z_N}{N^4} \leq \frac{64\pi}{N^5}.$$

The positive artanh series in (3) gives

$$|F_N(\beta) - N\lambda Z_N| \leq \frac{2|\beta|^3}{3(1-r^2)} \sum_{k\geq N} \mu_{N,k}^{-3} \leq \frac{64\pi|\lambda|^3}{9N^2}. \tag{18}$$

The coefficients d_k of the hypergeometric series in (15) are positive, with $d_1 = 1/16$ and $d_{k+1} < d_k$. Hence, for $|t| < 1/2$,

$$|\eta(t) - \pi t| \leq \frac{\pi|t|^3}{4(1-4t^2)}, \qquad |N\eta(\lambda/N) - \pi\lambda| \leq \frac{4\pi|\lambda|^3}{15N^2}.$$

Since $\varepsilon = -F_N(\beta)$, these inequalities imply

$$|e - \lambda(\pi - NZ_N)| \leq \frac{8\pi|\lambda|^3}{N^2}. \tag{19}$$

Equation (13) gives $\pi - NZ_N = \pi/(4N^2) + O(N^{-4})$. This and (19) prove the first estimate in (16); (12) and (18) prove the second. Dividing is legitimate: (3) is strictly increasing and vanishes only at $\beta = 0$. No differentiation of an unproved spectral asymptotic has been used. □

5 The actual central pair satisfies the hypothesis

Lemma 7 (An elementary central-mode bound)**.** *Let $N = 2m$, $m \geq 1$, and let $\lambda_{\min,+}(J_{2m})$ be the smallest positive eigenvalue. With $H_0 = 0$,*

$$0 < \lambda_{\min,+}(J_{2m}) \leq \frac{2m-1}{2m}\left(1 + \frac{1}{4}H_{m-1}\right)^{-1/2}. \tag{20}$$

In particular the innermost positive and negative modes satisfy (16) *and* (17)*.*

Proof. Use site indices $1, \ldots, 2m$ and bonds $b_j = j(2m-j)/(2m)$. Set $v_{2k} = 0$, $v_{2k-1} = A_k$, $A_1 = 1$, and $A_{k+1} = -b_{2k-1}A_k/b_{2k}$. All rows of $J_{2m}v$ vanish except the last, so $\|J_{2m}v\| = b_{2m-1}|A_m|$. With $c_r = (1/2)_r/r!$ as above,

$$|A_k| = \prod_{j=1}^{k-1} \frac{2j-1}{2j}\frac{2m-2j+1}{2m-2j} \geq c_{k-1}, \qquad |A_m| = (2m-1)c_{m-1}^2 \leq 1.$$

For the last inequality, the ratio of successive endpoint amplitudes is $1-1/(4m^2) < 1$, starting at one. Also rc_r^2 increases for $r \geq 1$, starting at $1/4$, because its successive ratio is $(2r+1)^2/[4r(r+1)] > 1$. Thus $\|v\|^2 \geq 1 + H_{m-1}/4$. Expanding v in an orthonormal eigenbasis gives $\operatorname{dist}(0, \operatorname{spec} J_{2m}) \leq \|J_{2m}v\|/\|v\|$. Finally the spectrum is symmetric, and $\det J_{2m} = (-1)^m \prod_{k=1}^{m} b_{2k-1}^2 \neq 0$. The distance is therefore the smallest positive eigenvalue, proving (20). The harmonic sum diverges, so the required nonzero eigenvalues tend to zero. □

The estimate is sufficient but is not claimed to give the sharp central eigenvalue scale. In combination with the previous theorem it gives the conservative central-pair bound $-4N^2 e_N/\varepsilon_N = 1 + O(1/\log N)$.

6 Independent recurrence and scope

An additional exact characterization, useful for checking constants, is

$$(N+1)^3 Z_{N+2} - 2N(N^2+1)Z_N + (N-1)^3 Z_{N-2} = 0, \qquad N \geq 4 \text{ even.} \tag{21}$$

For a direct derivation from (4), let $I_\nu = \int_{-1}^{1} (1-x^2)^{-1/2} P_\nu(x) P_\nu(-x) \mathrm{d}x$ and insert an extra x^2 to define T_ν. The degree raising and lowering identities [3], followed by two integrations by parts, give

$$I_{\nu+1} = \frac{\nu}{(\nu+1)^2} T_\nu + \left[-1 + \frac{1}{2(\nu+1)^2}\right] I_\nu,$$
$$I_{\nu-1} = -\frac{\nu+1}{\nu^2} T_\nu + \left[-1 + \frac{1}{2\nu^2}\right] I_\nu.$$

Endpoint products are $O(r^{1/2}(1+|\log r|))$ and vanish. Eliminating T_ν gives

$$(\nu+1)^3 I_{\nu+1} + (2\nu+1)(\nu^2+\nu+1/2) I_\nu + \nu^3 I_{\nu-1} = 0.$$

Substitution in (4), including the alternating sine factor, proves (21). The recurrence alone would not have fixed the normalization of a candidate solution; the contour argument supplies that missing information.

Numerical evaluation of the connection derivative and both positive-integral forms agrees at $N = 2, 4, 8$ to 70-digit working precision. Exact rational checks verify the displayed kernel coefficients and trial-vector identities. These calculations support reproducibility; the proofs and remainder estimates above do not depend on finite numerical agreement.

The conclusions concern even size and half filling. They prove a reciprocal-trace expansion and the signed law for shrinking Jacobi modes, including the central pair. They do not establish the uniform bulk quantum-period remainder, its higher spectral coefficients, a general-filling effective-size convention, or the full Wilson relative-entropy finite-part asymptotic. The separate companion *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law* proves the fixed-interior period expansion and leading Wilson law with bounded remainder.

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Positive Moment Bounds and Growing Plunge Windows

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Abstract

The reciprocal trace associated with the even half-filled sine kernel has a positive Stieltjes representation. It follows that every finite inverse-power expansion has a remainder of the sign of, and smaller than, its first omitted term. Two rational bounds control the exact central slope. The full formal inverse-power series nevertheless has zero radius of convergence. An elementary variational construction controls growing central windows of the commuting Jacobi matrix. An exact Darboux deletion, with its singular endpoint domains proved explicitly, realizes the omitted spectrum as a positive differential operator. Elementary bracketing then supplies higher reciprocal moments and proves the signed plunge law uniformly for $|\lambda| \le L_N = o(N^{1/4})$. In particular a logarithmic number of actual central pairs obey the law with relative error $O(N^{-1/2})$. No bulk quantum-period remainder is assumed.

1 The finite problem and exact inputs

Let $N = 2m$ be a positive even integer. With site indices $1, \dots, N$, define

$$B_{jk} = \begin{cases} 1/2, & j = k, \\ \sin[\pi(j-k)/2]/[\pi(j-k)], & j \neq k, \end{cases} \qquad (J_N)_{j,j+1} = (J_N)_{j+1,j} = b_j := \frac{j(N-j)}{N}, \quad (J_N)_{jj} = 0.$$

The matrices commute and $0 < B < I$. Write their common eigenvalues as $J_N v = \lambda v$ and $Bv = \zeta v$. Let $H = NJ_N/2$, $\alpha = (N^2-1)/4$, and let $\mu_{N,k}$, in increasing order starting at $k = 0$, be the eigenvalues of the Friedrichs realization

$$A_N u = -(\cos x\, u')' - \alpha \cos x\, u, \qquad -\pi/2 < x < \pi/2.$$

The endpoint condition excludes the logarithmic solution. The exact spectral-tail and trace identities established in [1, 2] are

$$\{\mu_{N,0}, \dots, \mu_{N,N-1}\} = -\operatorname{spec} H, \qquad \mu_{N,N} > \|H\|, \qquad Z_N := \sum_{k \ge N} \mu_{N,k}^{-1}, \tag{1}$$

$$\log \frac{\zeta}{1-\zeta} = F_N(N\lambda/2), \qquad F_N(b) = 2 \sum_{k \ge N} \operatorname{artanh} \frac{b}{\mu_{N,k}}, \tag{2}$$

$$Z_N = \pi \int_0^\infty e^{-Nt} f(t) \mathrm{d}t, \qquad f(t) = P_{-1/2}(\cosh t)^2. \tag{3}$$

Here P is the real Legendre function on $[1, \infty)$. The first identity identifies precisely the modes removed from the differential spectrum; the second holds at the actual common finite modes. The normalization in (3) follows from a weighted Legendre resolvent and a Mellin reflection identity, rather than an asymptotic fit. These previous results are the exact inputs to this paper. The positive integral in (3) defines an interpolation for every real $N > 0$; equality with the specified spectral trace is used only for even integers.

2 A positive Fourier measure

Lemma 1 (Autocorrelation identity)**.** *Put* $g(u) = \operatorname{sech}(u)^{1/2}$ *and use the Fourier convention* $\widehat{g}(\omega) = \int_{\mathbb{R}} e^{-i\omega u} g(u)\mathrm{d}u$*. Then*

$$y(t) := P_{-1/2}(\cosh t) = \frac{1}{\pi}\int_{\mathbb{R}} g(u)g(u-t)\mathrm{d}u, \tag{4}$$

$$\widehat{g}(\omega) = \frac{|\Gamma(1/4+i\omega/2)|^2}{\sqrt{2\pi}}, \qquad y(t) = \int_{\mathbb{R}} e^{i\omega t} p(\omega)\mathrm{d}\omega, \tag{5}$$

$$p(\omega) = \frac{|\Gamma(1/4+i\omega/2)|^4}{4\pi^3}. \tag{6}$$

The function p *is an even, strictly positive probability density with moments of every order. Consequently* $f = y^2$ *is the Fourier transform of the probability density* $q = p * p$*.*

Proof. The classical angular representation [3, Eq. 14.12.7] gives

$$y(t) = \frac{1}{\pi}\int_0^{\pi} (\cosh t + \sinh t \cos\theta)^{-1/2}\mathrm{d}\theta.$$

For $t \geq 0$, substitute $\tan(\theta/2) = e^u$. Then $\mathrm{d}\theta = \operatorname{sech} u\,\mathrm{d}u$ and

$$\cosh t + \sinh t\cos\theta = \frac{\cosh(u-t)}{\cosh u},$$

which proves (4). Both sides are even in t, so it holds on the whole real line. Substitution $x = e^{2u}$ in the Fourier integral, followed by Euler's beta integral [3, Eq. 5.12.3], gives

$$\widehat{g}(\omega) = 2^{-1/2}\int_0^{\infty} x^{1/4-i\omega/2-1}(1+x)^{-1/2}\mathrm{d}x = \frac{\Gamma(1/4-i\omega/2)\Gamma(1/4+i\omega/2)}{\sqrt{2\pi}}.$$

The function g and all its derivatives decay exponentially; in particular g is a Schwartz function. Fourier inversion for its autocorrelation proves (5). Plancherel and $\int_{\mathbb{R}} g^2\mathrm{d}u = \pi$ give $\int p = 1$. Positivity follows because the gamma function has no zeros. The Schwartz property supplies every moment and justifies differentiation under the Fourier integral. Multiplication of the two absolutely convergent Fourier integrals gives the convolution statement. □

Let σ be the image of $q(\omega)\mathrm{d}\omega$ under $s = \omega^2$. It is a probability measure on $[0,\infty)$, has full support, and has moments $m_k = \int_0^{\infty} s^k \mathrm{d}\sigma(s)$ for all $k \geq 0$.

Theorem 2 (Stieltjes representation and signed remainder)**.** *For the interpolation in* (3)*, for every* $N > 0$*,*

$$\boxed{\frac{NZ_N}{\pi} = \int_0^{\infty} \frac{\mathrm{d}\sigma(s)}{1+s/N^2}.} \tag{7}$$

For every integer $K \geq 0$*,*

$$0 < (-1)^{K+1}\left[\frac{NZ_N}{\pi} - \sum_{k=0}^{K}\frac{(-1)^k m_k}{N^{2k}}\right] < \frac{m_{K+1}}{N^{2K+2}}. \tag{8}$$

The formal series in N^{-2} *has zero radius of convergence, while* (8) *proves its asymptotic validity to every fixed order.*

Proof. Insert the Fourier representation for f in (3). Absolute Fubini is justified by $\int_{\mathbb{R}} q(\omega)\int_0^{\infty} Ne^{-Nt}\mathrm{d}t\,\mathrm{d}\omega = 1$. Integrating in t and using the evenness of q gives $N^2/(N^2+\omega^2)$, proving (7). The finite identity

$$\frac{1}{1+x} = \sum_{k=0}^{K}(-x)^k + \frac{(-1)^{K+1}x^{K+1}}{1+x}, \qquad x \geq 0,$$

then gives the signed remainder in (8) exactly as

$$\frac{1}{N^{2K+2}}\int_0^{\infty}\frac{s^{K+1}}{1+s/N^2}\mathrm{d}\sigma(s).$$

It is strictly between zero and m_{K+1}/N^{2K+2} because σ is not supported at zero. For every $R > 0$, $m_k \geq R^k \sigma([R, \infty))$, with a strictly positive last factor. Therefore $m_k^{1/k} \to \infty$, proving zero convergence radius. The estimate at each fixed K already proves the asymptotic assertion. □

Since $f^{(2k)}(0) = (-1)^k m_k$, its differential equation or hypergeometric series determines the moments exactly:

$$\begin{array}{c|cccccc} k & 0 & 1 & 2 & 3 & 4 & 5 \\ \hline m_k & 1 & \frac{1}{4} & \frac{17}{64} & \frac{169}{256} & \frac{50777}{16384} & \frac{1575169}{65536}. \end{array} \tag{9}$$

For example, if

$$S_N = \frac{\pi}{N} - \frac{\pi}{4N^3} + \frac{17\pi}{64N^5} - \frac{169\pi}{256N^7} + \frac{50777\pi}{16384N^9},$$

then for every positive even N,

$$0 < S_N - Z_N < \frac{1575169\pi}{65536N^{11}}. \tag{10}$$

These are bounds on the exact trace, not just on formal coefficients.

Corollary 3 (Two rational slope bounds)**.** *For every positive even N,*

$$\boxed{\frac{4\pi}{16N^2 + 17} < \pi - NZ_N < \frac{\pi}{4N^2 + 1}.} \tag{11}$$

Also

$$\left|\pi - NZ_N - \frac{\pi}{4N^2}\right| < \frac{17\pi}{64N^4}. \tag{12}$$

Proof. Strict Jensen applied to $s \mapsto (1 + s/N^2)^{-1}$ under σ gives $NZ_N/\pi > (1 + m_1/N^2)^{-1}$. For the probability measure $\mathrm{d}\tau = s\mathrm{d}\sigma/m_1$, its mean is $m_2/m_1 = 17/16$ and

$$1 - \frac{NZ_N}{\pi} = \frac{m_1}{N^2} \int \frac{\mathrm{d}\tau(s)}{1 + s/N^2} > \frac{m_1}{N^2 + m_2/m_1}.$$

Both measures are nondegenerate, so both inequalities are strict. Substitution of the moments proves (11); the case $K = 1$ of (8) proves (12). □

3 A growing window of small Jacobi eigenvalues

List the positive eigenvalues of J_{2m} increasingly as $0 < \lambda_{1,+} \leq \cdots \leq \lambda_{m,+}$. The negative eigenvalues are their opposites: conjugation by alternating signs sends J to $-J$. There is no zero eigenvalue, since $\det J_{2m} = (-1)^m \prod_{k=1}^m b_{2k-1}^2 \neq 0$.

Theorem 4 (Explicit logarithmic-window bound)**.** *Set $L = \lfloor m/2 \rfloor$. For any integer $r \geq 1$ with $16^r \leq L$, put $\ell = \lfloor \log L/(r \log 16) \rfloor$. Then*

$$\lambda_{r,+} \leq \frac{12\sqrt{2}\,\pi}{\ell \log 16}. \tag{13}$$

In particular, if $1 \leq r \leq \log L/(2 \log 16)$, then

$$\boxed{\lambda_{r,+} \leq \frac{24\sqrt{2}\,\pi r}{\log L}.} \tag{14}$$

Thus the innermost r_N pairs shrink uniformly to zero whenever $r_N = o(\log N)$ as even $N \to \infty$.

Proof. Define $A_1 = 1$ and $A_{k+1} = -b_{2k-1}A_k/b_{2k}$ for $k < m$. For $v_{2k-1} = A_k f_k$, $v_{2k} = 0$, direct substitution gives

$$\|v\|^2 = \sum_{k=1}^m A_k^2 f_k^2, \qquad \|Jv\|^2 = \sum_{k=1}^m (b_{2k-1}A_k)^2 (f_k - f_{k+1})^2, \tag{15}$$

where $f_{m+1} = 0$ encodes the actual last row. For $1 \le k \le L$,

$$\frac{1}{4k} \le A_k^2 \le \frac{3}{k}, \qquad (b_{2k-1}A_k)^2 \le 12k. \tag{16}$$

Indeed, with $c_s = (1/2)_s/s!$,

$$|A_k| = c_{k-1} \prod_{j=1}^{k-1} \left(1 + \frac{1}{2(m-j)}\right).$$

For $s \ge 1$, sc_s^2 increases from $1/4$, because its successive ratio is $(2s+1)^2/[4s(s+1)] > 1$. Also $(s+1)c_s^2$ decreases from one, because its successive ratio is $1 - (3s+2)/[4(s+1)^3] < 1$. These prove the required lower bound and $c_{k-1}^2 \le 1/k$. For $k \le L$, the logarithm of the squared product is at most

$$\sum_{j=1}^{k-1} \frac{1}{m-j} \le \frac{k-1}{m-k+1} < 1.$$

The squared product is therefore below $e < 3$. Checking $k = 1$ directly and using $b_{2k-1} \le 2k$ proves (16).

Set $R = 16^\ell$ and $h = \log R$. On each integer block $[a, b] = [R^j, R^{j+1}]$, $j = 0, \dots, r-1$, use the sequence

$$f_k = \begin{cases} \sin[\pi \log(k/a)/h], & a \le k \le b, \\ 0, & \text{otherwise.} \end{cases}$$

All blocks lie in $[1, L]$, and their shared endpoints are zero. The resulting odd-site vectors have disjoint nonzero supports; so do their images under J, since a block contributes only to the even rows corresponding to $k = a, \dots, b-1$ in (15). The norm and energy forms consequently have no cross terms between different packets.

The numbers $aR^{1/4}$ and $aR^{3/4}$ are integers, and $f_k^2 \ge 1/2$ between them. Thus

$$\|v\|^2 \ge \frac{1}{8} \sum_{k=aR^{1/4}}^{aR^{3/4}} \frac{1}{k} \ge \frac{1}{8} \int_{aR^{1/4}}^{aR^{3/4}+1} \frac{\mathrm{d}x}{x} \ge \frac{h}{16}.$$

The sine function is 1-Lipschitz and $\log(1 + 1/k) \le 1/k$, so $|f_{k+1} - f_k| \le \pi/(hk)$ inside the block. Therefore

$$\|Jv\|^2 \le \frac{12\pi^2}{h^2} \sum_{k=a}^{b-1} \frac{1}{k} \le \frac{12\pi^2(h+1)}{h^2} \le \frac{18\pi^2}{h},$$

where $h \ge \log 16 > 2$. Every nonzero vector in the r-dimensional packet span thus has $\|Jv\|^2/\|v\|^2 \le 288\pi^2/h^2$.

Order the odd sites before the even sites. Then

$$J = \begin{pmatrix} 0 & T^{\mathsf{T}} \\ T & 0 \end{pmatrix},$$

where T is invertible and upper bidiagonal, with diagonal $b_1, b_3, \dots, b_{2m-1}$. Its singular values are precisely the positive eigenvalues of J. Min–max on the odd-site block $T^{\mathsf{T}}T$ therefore gives $\lambda_{r,+}^2 \le 288\pi^2/h^2$, proving (13). If $\log L/(r \log 16) \ge 2$, the floor is at least half its argument; this proves (14). Finally $\log L \sim \log N$. □

The constants are deliberately elementary, not optimal. The finite-size condition for (13) already requires $N \ge 64$ when $r = 1$; the assertion is not applied outside its stated range.

4 Uniform transfer to the modular error

For a common mode define

$$\eta(t) = \pi t \, {}_3F_2(1/4, 1/2, 3/4; 1, 3/2; 4t^2), \qquad \varepsilon = \log \frac{1-\zeta}{\zeta}, \qquad e = \varepsilon + N\eta(\lambda/N).$$

All quantities are dimensionless. In particular the word “modular” here does not supply a physical clock or an energy unit.

Lemma 5 (Explicit ratio control)**.** *For every even $N \geq 8$ and every common mode with $0 < |\lambda| \leq 1$,*

$$\left| \frac{-4N^2 e}{\varepsilon} - 1 \right| \leq \frac{10}{N^2} + 33\lambda^2. \tag{17}$$

Proof. The two central sites give $\|H\| \geq N^2/8$ by a Rayleigh quotient. From (1), with $\beta = N\lambda/2$,

$$\frac{|\beta|}{\mu_{N,N}} \leq \frac{4|\lambda|}{N} \leq \frac{1}{2}, \qquad \sum_{k \geq N} \mu_{N,k}^{-3} \leq \frac{64 Z_N}{N^4} \leq \frac{64\pi}{N^5}.$$

The positive artanh series therefore yields

$$|F_N(\beta) - N\lambda Z_N| \leq \frac{64\pi|\lambda|^3}{9N^2}. \tag{18}$$

The coefficients d_k of the hypergeometric series in the definition of η are positive and decreasing, with $d_1 = 1/16$. Indeed their successive ratio is $[(k+1/4)(k+1/2)(k+3/4)]/[(k+1)^2(k+3/2)] < 1$. Thus

$$|N\eta(\lambda/N) - \pi\lambda| \leq \frac{\pi|\lambda|^3}{4N^2(1 - 4\lambda^2/N^2)} \leq \frac{4\pi|\lambda|^3}{15N^2}.$$

Since $\varepsilon = -F_N(\beta)$, combining this with (18) gives

$$|e - \lambda(\pi - NZ_N)| \leq \frac{8\pi|\lambda|^3}{N^2}.$$

Set $a = 4N^2 e/(\pi\lambda)$ and $b = F_N(\beta)/(\pi\lambda)$. Equations (12) and (18), including the common sign of each artanh term and λ, imply

$$|a - 1| \leq \frac{17}{16N^2} + 32\lambda^2, \qquad b \geq 1 - \frac{1}{4N^2} > 0, \qquad |b - 1| \leq \frac{8}{N^2}.$$

Consequently

$$\left| \frac{a}{b} - 1 \right| \leq \frac{145/(16N^2) + 32\lambda^2}{1 - 1/(4N^2)} \leq \frac{10}{N^2} + 33\lambda^2.$$

Here $a/b = -4N^2 e/\varepsilon$. Strict positivity of b also justifies division by ε. □

Corollary 6 (Uniform signed plunge law on a growing window)**.** *Let $r_N \geq 1$ be integers with $r_N = o(\log N)$ as even $N \to \infty$. For both signs of the innermost r_N pairs of common eigenmodes,*

$$\sup_{1 \leq j \leq r_N,\, \pm} \left| \frac{-4N^2 e_{j,\pm}}{\varepsilon_{j,\pm}} - 1 \right| = O\left(N^{-2} + \frac{r_N^2}{\log^2 N} \right) \longrightarrow 0. \tag{19}$$

For all sufficiently large N, an explicit upper bound for the left side is

$$\frac{10}{N^2} + 33 \left(\frac{24\sqrt{2}\,\pi r_N}{\log\lfloor N/4 \rfloor} \right)^2.$$

Proof. The bound (14) puts every eigenvalue in the window below one in absolute value for all sufficiently large N. Apply (17) to each mode. Its constants are uniform, so taking the supremum preserves the estimate. □

In particular every fixed number of central pairs obeys the law, with the conservative relative error $O((\log N)^{-2})$.

5 An exact positive operator for the differential tail

The higher reciprocal moments admit a useful positive realization. Write $p(x) = \cos x$ and introduce the Liouville coordinate

$$z(x) = \int_{-\pi/2}^{x} \frac{\mathrm{d}u}{\sqrt{\cos u}}, \qquad 0 < z < \mathcal{L}, \qquad \mathcal{L} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{\Gamma(3/4)}.$$

Below $p(z) = \cos x(z)$.

Theorem 7 (Exact deletion of the finite block)**.** *The spectrum of*

$$\boxed{T_N = -D_z^2 + \frac{4N^2-1}{16p(z)} + \frac{p(z)}{16}} \tag{20}$$

with its Friedrichs realization is precisely $\{\mu_{N,k} : k \geq N\}$*. Its form domain is* $H_0^1(0,\mathcal{L})$*, and its endpoints are limit point. In particular,*

$$\mu_{N,N} > N^2/4, \qquad Z_N(s) := \sum_{k\geq N} \mu_{N,k}^{-s} = \operatorname{Tr} T_N^{-s} \quad (s > 1/2). \tag{21}$$

Proof. The unitary map $\psi(z) = p(x(z))^{1/4}u(x(z))$ transforms A_N into $-D_z^2 + V_N$, where direct differentiation gives

$$V_N = -N^2 p/4 + p/16 - 1/(16p).$$

At either endpoint, with d the distance to that endpoint, $p = d^2/4 + O(d^6)$. Thus $V_N = -1/(4d^2) + O(d^2)$, and the selected eigenfunctions behave as $d^{1/2}$ times a regular power series in d^2. The logarithmic branch is excluded. This critical endpoint requires a domain argument in addition to the usual formal Darboux–Crum transformation [4].

For one deletion, let $g > 0$ be the current ground eigenfunction, with eigenvalue E_0, and suppose its selected endpoint exponent is $k + 1/2$, $k \geq 0$. Define Q as the closure from C_c^∞ of $D_z - g'/g$. The compact-support ground-state identity, followed by closure of quadratic forms, gives

$$S - E_0 = Q^*Q$$

for the current Friedrichs operator S. The self-adjoint partner $QQ^* + E_0$ extends the differential expression with potential $V - 2(\log g)''$. Differentiable Frobenius expansions show that its endpoint coefficient is $((k+1)^2 - 1/4)d^{-2} + O(1)$. Its two branches have exponents $k + 3/2$ and $-k - 1/2$; only the first is square integrable. Hence the partner is limit point at both endpoints and is its unique self-adjoint realization. Moreover $\ker Q^* = \{0\}$, since $Q^*f = 0$ implies $f = c/g$, which is not square integrable, including at $k = 0$. The gap above E_0 makes $\operatorname{ran} Q$ closed; its orthogonal complement is $\ker Q^*$, so it is all of L^2. Polar decomposition therefore pairs the orthogonal complement of g unitarily with the partner space. Exactly the ground eigenvalue is deleted, with completeness and compact resolvent preserved. The new ground state has exponent $k + 3/2$, permitting iteration.

Direct finite Fourier transformation and (1) show that the first N original eigenfunctions span

$$\operatorname{span}\{e^{imx} : m = -(N-1)/2, \ldots, (N-1)/2\}.$$

Their transformed Wronskian is therefore

$$W_z(\psi_0, \ldots, \psi_{N-1}) = C\, p^{N^2/4}, \qquad C \neq 0.$$

Indeed the common factor $p^{1/4}$ contributes $p^{N/4}$, the change of variable contributes $(\mathrm{d}x/\mathrm{d}z)^{N(N-1)/2} = p^{N(N-1)/4}$, and the Fourier Wronskian is a nonzero constant Vandermonde determinant. Iterating the one-step identity gives $V_{\text{tail}} = V_N - 2D_z^2 \log|W_z|$. Since

$$D_z^2 \log p = -\tfrac{1}{2}(p + p^{-1}),$$

this is exactly (20). The Wronskian formula evaluates the already justified self-adjoint deletions; it does not replace their domain argument.

The final potential is $(N^2 - 1/4)d^{-2} + O(1)$. Hardy's inequality gives form domain H_0^1. Finally

$$\frac{4N^2-1}{16p} + \frac{p}{16} - \frac{N^2}{4} = \frac{(1-p)(4N^2-1-p)}{16p} \geq 0.$$

The kinetic energy is strictly positive on H_0^1, proving the first assertion of (21). The trace convergence and its quantitative form also follow from the bracketing below. □

6 Higher reciprocal moments by elementary bracketing

Theorem 8 (Higher-moment leading terms with an explicit error)**.** *For each fixed* $s > 1/2$*,*

$$Z_N(s) = N^{1-2s}\{I_s + O_s(N^{-1/2})\}, \qquad I_s = \frac{2^{2s-2}\Gamma(s-1/2)\Gamma(s/2)}{\Gamma(s)\Gamma((s+1)/2)}. \tag{22}$$

For every integer $M \geq 1$ one has, in particular,

$$\left|N^5 Z_N(3) - 3\pi\right| \leq D_{N,M} := \frac{6\mathcal{L}}{M} + \frac{128M}{N} + 3\pi\left[\left(1 - \frac{1}{4N^2}\right)^{-5/2} - 1\right]. \tag{23}$$

Proof. Put $h = 1/N$ and $Q_h = h^2 T_N = -h^2 D_z^2 + V_h$. Then

$$V_h = \left(1 - \frac{h^2}{4}\right) V_0 + \frac{h^2 p}{16}, \qquad V_0 = \frac{1}{4p}, \qquad (1 - h^2/4)V_0 \leq V_h \leq V_0.$$

The potential V_h has minimum $1/4$ and is monotone on each half-interval. Divide each half into M cells of length $\delta = \mathcal{L}/(2M)$. Neumann conditions and the cell infimum of V_h give an upper trace bound by the min-max principle. Dirichlet conditions and the cell supremum give a lower trace bound. The two exterior cells, whose supremum is infinite, are omitted from the latter bound; this is a restriction of variational subspaces, not a boundedness assumption at the singular endpoints.

For constant potential $v > 0$ the cell eigenvalues are $(h\pi j/\delta)^2 + v$, with $j \geq 0$ for Neumann and $j \geq 1$ for Dirichlet conditions. The integral test gives

$$\left|h \operatorname{Tr}(-h^2 D_{\mathrm{N/D}}^2 + v)^{-s} - c_s \delta v^{1/2-s}\right| \leq h v^{-s}, \qquad c_s = \frac{\Gamma(s - 1/2)}{2\sqrt{\pi}\,\Gamma(s)}.$$

The total cell trace error is at most $2Mh4^s$. The function $V_h^{1/2-s}$ rises from zero to $4^{s-1/2}$ and falls symmetrically; its upper and lower Riemann sums differ by at most $2\delta 4^{s-1/2}$. Therefore, with $I_s(h) = c_s \int_0^{\mathcal{L}} V_h^{1/2-s}\mathrm{d}z$,

$$\left|h \operatorname{Tr} Q_h^{-s} - I_s(h)\right| \leq \frac{c_s \mathcal{L} 4^{s-1/2}}{M} + 2Mh4^s.$$

This also proves convergence of the trace. Potential comparison gives $I_s \leq I_s(h) \leq (1 - h^2/4)^{-(s-1/2)} I_s$, where

$$I_s = c_s 4^{s-1/2} \int_{-\pi/2}^{\pi/2} (\cos x)^{s-1} \mathrm{d}x.$$

The beta integral evaluates this as (22). Since $h \operatorname{Tr} Q_h^{-s} = N^{2s-1} Z_N(s)$, choosing $M = \lceil \sqrt{N} \rceil$ proves the stated rate. For $s = 3$, $c_3 = 3/16$ and $I_3 = 3\pi$, giving (23). □

For example $I_5 = 105\pi/4$. The first-moment result obtained by this general argument is weaker than (3); the new information is the independently proved higher-moment coefficient. No bulk period expansion was used.

7 A larger uniform eigenvalue window

The third-moment coefficient cancels the leading nonlinear term of the modular error.

Theorem 9 (Uniform center law beyond shrinking eigenvalues)**.** *Let $L_N = o(N^{1/4})$ as even $N \to \infty$. Uniformly over nonzero common modes with $|\lambda| \leq L_N$,*

$$e = \frac{\pi\lambda}{4N^2}\left[1 + O\left(N^{-2} + \lambda^2 N^{-1/2} + \lambda^4 N^{-2}\right)\right], \tag{24}$$

$$\frac{-4N^2 e}{\varepsilon} \longrightarrow 1. \tag{25}$$

This includes every fixed bounded eigenvalue window. More explicitly, for even $N \geq 2$, $0 < |\lambda| \leq N/4$, and any integer $M \geq 1$,

$$\left|\frac{-4N^2 e}{\varepsilon} - 1\right| \leq \frac{\dfrac{21}{16N^2} + \left(\dfrac{D_{N,M}}{3\pi} + \dfrac{16}{9N^2}\right)\lambda^2 + \dfrac{20\lambda^4}{N^2}}{1 - \dfrac{1}{4N^2}}. \tag{26}$$

Proof. Set $\beta = N\lambda/2$. By (21), $r = |\beta|/\mu_{N,N} < 2|\lambda|/N \le 1/2$. The exact artanh series (2) gives

$$F_N(\beta) = N\lambda Z_N + \frac{N^3\lambda^3}{12} Z_N(3) + R, \qquad |R| \le \frac{2|\beta|^5}{5(1-r^2)} Z_N(5).$$

The simple bound $Z_N(5) \le (4/N^2)^4 Z_N \le 256\pi/N^9$ suffices, giving $|R| \le 64\pi|\lambda|^5/(15N^4)$. The convergent hypergeometric series defining η, with positive decreasing coefficients, gives

$$N\eta(\lambda/N) = \pi\lambda + \frac{\pi\lambda^3}{4N^2} + R_\eta, \qquad |R_\eta| \le \frac{7\pi|\lambda|^5}{16N^4}.$$

Equation (23) cancels the two cubic terms, and (12) therefore implies

$$\left| e - \frac{\pi\lambda}{4N^2} \right| \le \frac{17\pi|\lambda|}{64N^4} + \frac{D_{N,M}|\lambda|^3}{12N^2} + \frac{5\pi|\lambda|^5}{N^4}.$$

Choosing $M = \lceil\sqrt{N}\rceil$ proves (24). For division, put $a = 4N^2 e/(\pi\lambda)$ and $b = F_N(\beta)/(\pi\lambda)$. The sign of every artanh term gives $b \ge NZ_N/\pi \ge 1 - 1/(4N^2) > 0$. Also $Z_N(3) \le 16Z_N/N^4$ and the artanh bound before the cubic term give

$$|b-1| \le \frac{1/4 + 16\lambda^2/9}{N^2}.$$

Combining this with the preceding bound on $a-1$ gives (26), since $a/b = -4N^2 e/\varepsilon$. On the stated window all its terms tend uniformly to zero, proving (25). □

Corollary 10 (A logarithmic number of actual central pairs)**.** *For all sufficiently large even N, put*

$$r_N = \left\lfloor \frac{\log\lfloor N/4\rfloor}{\log 16} \right\rfloor.$$

For both signs of the innermost r_N pairs of common modes,

$$\sup_{1\le j\le r_N,\,\pm} \left| \frac{-4N^2 e_{j,\pm}}{\varepsilon_{j,\pm}} - 1 \right| = O(N^{-1/2}).$$

Proof. The choice of r_N satisfies $16^{r_N} \le \lfloor N/4\rfloor$. The integer ℓ in (13) is at least one, so

$$|\lambda_{j,\pm}| \le \frac{12\sqrt{2}\,\pi}{\log 16} \qquad (1 \le j \le r_N).$$

Apply (26) on this fixed bounded window with $M = \lceil\sqrt{N}\rceil$. Its constants are independent of the selected mode. The numerical constant and the bracketing rate are not asserted to be optimal. □

This is still a center result: $t = \lambda/N = o(N^{-3/4})$. The argument proves no fixed-t bulk quantum-period remainder and does not claim the displayed bracketing rate or window exponent is optimal.

8 Literature comparison and scope

The logarithmic scale has a rigorous precedent. Deift, Its, and Krasovsky [5, Theorem 1.3] prove that, for a symbol taking two positive constant values on complementary rational arcs, eigenvalues in a fixed compact subinterval of the gap have spacing of order $1/\log N$; every interval of a suitable constant times $1/\log N$ contains an eigenvalue. The choice of values $1, 2$ and a half-circle gives $I + B_N$ up to diagonal unitary conjugation. This supports the known logarithmic occupation scale. The finite Jacobi construction above supplies an independent bound with explicit constants. No new scale or priority is claimed.

The positive-measure theorem controls the exact first reciprocal trace at every even size. The growing-window result concerns actual common eigenmodes, and its proof is uniform over the stated window. It does not establish a uniform bulk quantum-period expansion, an odd-size deflation identity, or a general-filling effective-size convention. The fixed-interior bulk expansion is supplied separately by *Uniform Bulk Modular Asymptotics and the Leading Wilson Entropy Law*; it is not a consequence of moment bracketing alone. In particular, the trace measure's

moments m_k are not the higher reciprocal spectral traces $\sum_{j\geq N} \mu_{N,j}^{-(2k+1)}$. Their leading asymptotics are established separately by (22).

Exact arithmetic checks recover the displayed moments independently from the hypergeometric series. Quadrature checks compare the autocorrelation, gamma density, and Legendre representations, and finite matrix checks verify the packet energies. Symbolic checks also verify the transformed potential, Wronskian exponent, and moment constants. These checks provide reproducibility; the analytic inequalities and min–max argument are the proofs. No finite numerical agreement is used to infer a theorem.

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